

Gathetr Owner's Technical Operations, Development, Maintenance and Repair Manual

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Manual

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Document Control

This manual covers the runtime
application source at the recorded
application commit and the
documentation source at the recorded
documentation commit. It does not
claim to describe later changes unless
regenerated.

Source documents in `docs/` are
authoritative. The HTML and PDF
outputs are generated reading copies.

How to Use This Manual

- Start with the Emergency
Reference when the app is
failing right now.
- Use the main manual
chapters for architecture,
routes, permissions, and
feature maintenance.
- Use the Runbook Section for
step-by-step diagnosis and
repair.
- Use the appendices for field-
level Firestore details,
command safety, and
route/file indexes.

Revision History

- Phase 3: initial technical summary and first generated PDF.
- Phase 3B: rendering repair, runbook expansion, appendices, bookmarks, and PDF validation.

Protected Boundaries

- Do not expose `.env.local`, cookies, tokens, service account JSON, or private Firebase keys.
- Do not use CPB for synthetic QA or fake data.
- QR payload format must remain `GSV:TICKET:{ticketCode}` unless explicitly approved and migrated.
- Firestore Rules, Auth, Hosting, Functions, and production data require explicit validation and approval before deployment.
- This Phase 3B documentation generation requires no Firebase deployment and makes no runtime behavior change.

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Quick Start and Emergency Reference

Immediate Symptom Table

Symptom	First check	Next manual section
Blank page	Browser Console and Network; check stale chunk or Firebase config	Troubleshooting: blank page or dynamic import failure
Permission denied	Signed-in account, settings/accessControl, staff assignment, Firestore rule line	Permissions and Security; Permission runbooks
Login fails	Firebase Auth provider, authorized domain, popup/redirect state	Firebase Authentication; Login failure runbook
Scanner fails	Camera permission, HTTPS, QR payload, assigned event, ticket code lookup	Imports/Exports/QR; Scanner runbook
Build fails	Terminal error and changed files	Build Deployment and Recovery
Production fails	Hosting deployment, asset cache, Firebase project, Console	Emergency recovery runbooks

Safe First Commands

```
git status --short --branch
git rev-list --left-right --count
main...origin/main
npm run lint
npm test
npm run build
```

Do not run deployment commands while diagnosing unless the fix is known, tested, and approved.

Application Overview

Gathetr is a private React/Firebase event operations application for planning, guest management, registration payments, tickets, check-in, operations, run of show, resources, documents, contacts, reports, imports, message preparation, Settings, and System QA.

The active synthetic training event is CODEX_DEMO - Full System Walkthrough. CPB is real historical production data and must not receive synthetic writes.

Main User Groups

- Protected Owner: UID-protected permanent owner access.
- Approved Organizer: active approved email in settings/accessControl.
- Staff roles: active staffProfiles/{uid} plus active event assignment.
- Scanner: assigned-event check-in-only route and narrow Firestore update path.

Stack

Layer	Actual technology
Frontend	React ^19.2.0, Vite ^7.2.4, React Router ^8.3.0
Styling	Tailwind CSS ^4.1.17 through @tailwindcss/vite
Backend services	Firebase client SDK ^12.6.0
Database	Cloud Firestore with local emulator tests
Authentication	Firebase Authentication with Google and email/password
Hosting	Firebase Hosting classic static SPA rewrites to index.html
QR	qrcode ^1.5.4; scanner uses html5-qrcode ^2.3.8
Excel parsing	read-excel-file ^9.2.0; xlsx package intentionally absent

System Architecture

High-Level Flow



Caption: high level architecture diagram.

Important Boundaries

- Frontend route checks improve user experience but do not replace Firestore Rules.
- Firestore Rules are the backend security boundary.
- Audit logs are append-only evidence and are coupled to business mutations.
- Message Builder is copy-only; it does not send email.
- Documents are references/links and metadata only; no Firestore Storage upload is active.

Project File Map

Feature Ownership Map

Feature	Main files	Related tests	Risk
Authentication and access resolution	src/lib/firebase.js; src/auth/AuthProvider.jsx; src/auth/ProtectedRoute.jsx; src/utls/accessRoles.js; src/config/protectedOwner.js	tests/auth-reliability.test.js; tests/protected-owner-authorization-matrix.test.js; tests/settings-access-management.test.js	High
Events and working event	src/pages/EventsPage.jsx; src/services/eventService.js; src/events/ActiveEventProvider.jsx; src/utls/eventPlanning.js	tests/event-utls.test.js; tests/phase26-event-categories-capabilities.test.js	High
Tasks and deadlines	src/pages/TasksPage.jsx; src/services/taskService.js; src/utls/taskWorkflow.js	tests/phase2-guided-event-setup-task-deadline.test.js; tests/task-workflow-rules.test.js	Medium
Guests and registrations	src/pages/RegistrationsPage.jsx; src/services/registrationService.js; src/utls/validators.js; src/utls/registrationMetrics.js	tests/registration-utls.test.js; tests/registration-metrics.test.js	High
Registration payments	src/pages/PaymentsPage.jsx; src/utls/financeUtils.js; src/utls/paymentStatus.js	tests/phase23b-payments-operations-boundaries.test.js; tests/payments-operations-refinement-2026-08.test.js	High
Review and reconciliation	src/pages/PaymentReconciliationPage.jsx; src/services/reconciliationReadService.js; src/utls/paymentReconciliation.js; src/utls/reconciliationWorkbook.js	tests/phase23c-payment-reconciliation.test.js	High
Tickets and QR	src/pages/TicketsPage.jsx; src/components/tickets/TicketQrCode.jsx; src/services/ticketService.js; src/utls/ticketUtils.js; src/utls/qrTicketUtils.js	tests/phase45-ticketing.test.js; tests/phase7-qr-checkin.test.js	High
Check-in and scanner	src/pages/CheckInPage.jsx; src/pages/ScannerPage.jsx; src/components/checkin/QrScannerPanel.jsx; src/utls/checkInUtils.js	tests/phase14-camera-checkin.test.js; tests/firestore-checkin-rules.test.js; e2e/workflows.spec.js	High
Imports and XLSX parsing	src/pages/ImportsPage.jsx; src/services/importService.js; src/utls/importUtils.js; src/utls/xlsxImport.js	tests/import-center.test.js; tests/import-center-workflow-upgrade.test.js	High
Operations ledger and commitments	src/pages/OperationsPage.jsx; src/services/operationsLedgerService.js; src/components/operations/PartnerCommitmentsPanel.jsx	tests/phase19-operations-productivity.test.js; tests/phase23b-payments-operations-boundaries.test.js	High
Run of Show	src/pages/RunOfShowPage.jsx; src/services/runOfShowService.js; src/utls/runOfShow.js	tests/run-of-show-resources-foundation.test.js; tests/run-of-show-resource-rules.test.js	Medium
Equipment and supplies	src/pages/ResourcesPage.jsx; src/services/eventResourceService.js; src/utls/eventResources.js	tests/run-of-show-resources-foundation.test.js;	Medium

Feature	Main files	Related tests	Risk
		tests/run-of-show-resource-rules.test.js	
Documents	src/pages/DocumentsPage.jsx; src/services/documentService.js; src/utills/documentRegistry.js	tests/document-contact-foundation.test.js; tests/document-contact-rules.test.js	Medium
Contacts and organizations	src/pages/ContactsPage.jsx; src/services/contactService.js; src/utills/contactDirectory.js	tests/document-contact-foundation.test.js; tests/document-contact-rules.test.js	Medium
Settings, staff, integrations	src/pages/SettingsPage.jsx; src/services/accessManagementService.js; src/services/staffManagementService.js; src/services/integrationSettingsService.js	tests/settings-access-management.test.js	High
Message Builder	src/pages/CommunicationsPage.jsx; src/utills/communicationsUtils.js	tests/phase6-communications.test.js	Medium
System QA	src/pages/QaPage.jsx; src/utills/qaHelper.js; src/utills/runtimeHealth.js	tests/production-qa.test.js; tests/settings-systemqa-tutorial-final-refinement.test.js	Medium
Tutorial/onboarding	src/tutorial/*	tests/onboarding-flow.test.js; tests/phase26-interactive-product-tour.test.js; e2e/tutorial.spec.js	Medium

Root Configuration

Path	Purpose	Risk
package.json	Scripts and dependencies for development, QA, E2E, Firebase deploy helpers, docs generation.	High
firebase.json	Auth provider notes, emulator ports, Firestore file references, Hosting headers and rewrites.	High
firestore.rules	Backend authorization and schema validation.	Critical
firestore.indexes.json	Firestore composite index configuration.	High
vite.config.js	Vite/React/Tailwind build configuration.	Medium
playwright.config.js	E2E browser test configuration and emulator env.	Medium
AI_AGENT_RULES.md	Permanent coding assistant governance.	High

Frontend React and Tailwind

Entry point: `src/main.jsx` Top-level app: `src/App.jsx` Authenticated shell: `src/layout/AppShell.jsx` Global styles: `src/styles.css`

React Patterns

- Lazy-loaded route pages are declared in `src/App.jsx`.
- `ProtectedRoute` blocks unauthenticated or unauthorized users.
- `AppShell` owns desktop sidebar, mobile drawer, mobile tab bar, page titles, Admin Search, page guidance, and `TutorialProvider` wrapping.
- Firebase writes are kept in service modules under `src/services/`.
- Shared validation and display helpers live in `src/utis/`.

Tailwind Usage

Tailwind is used directly through class names and the Vite Tailwind plugin. The app uses compact cards, dense grids, responsive `sm`, `md`, `lg`, `xl` breakpoints, and semantic color usage tied to the Gather & Savor visual system.

Troubleshooting:

- Element not visible: check responsive utility prefixes and route/access gating.
- Mobile overflow: inspect fixed widths, tables, and long event names.
- Dynamic class not applying: avoid constructing class strings that Tailwind cannot see statically.
- Form accessibility issue: inspect associated label, id, name, and focus behavior.

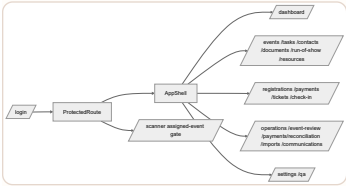
Routing and Navigation

Routes are declared in `src/App.jsx`;
page titles and navigation groups are in
`src/layout/AppShell.jsx`.

Route Map

Route	Title	Purpose	Component/Gate	Access	Source
/dashboard	Event Overview	Current event status, priorities, and next actions		Authenticated route gate	src/App.jsx; src/layout/AppShell.jsx
/events	Events	Plan and organize every gathering		Authenticated route gate	src/App.jsx; src/layout/AppShell.jsx
/tasks	Tasks & Deadlines	Event-scoped work, blockers, and follow-up dates		Authenticated route gate	src/App.jsx; src/layout/AppShell.jsx
/registrations	Guests & Registrations	Manage registration records and guest counts		Authenticated route gate	src/App.jsx; src/layout/AppShell.jsx
/payments	Registration Payments	Review registration charges, payments, balances, and follow-up		Authenticated + assigned Working Event gate	src/App.jsx; src/layout/AppShell.jsx
/payments/reconciliation	Review & Reconcile Records	Read-only workbook comparison before any correction		Authenticated route gate	src/App.jsx; src/layout/AppShell.jsx
/tickets	Tickets	Assign ticket codes and prepare QR access		Authenticated route gate	src/App.jsx; src/layout/AppShell.jsx
/check-in	Check-In	Track event-day attendance		Authenticated + assigned Working Event gate	src/App.jsx; src/layout/AppShell.jsx
/operations	Operations	Track event-level money and obligations		Authenticated + assigned Working Event gate	src/App.jsx; src/layout/AppShell.jsx
/run-of-show	Run of Show	Event-day sequence, supplier arrivals, dependencies, and Now/Next		Authenticated + assigned Working Event gate	src/App.jsx; src/layout/AppShell.jsx
/resources	Equipment & Supplies	Equipment, supplies, packing, pickup, and return tracking		Authenticated + assigned Working Event gate	src/App.jsx; src/layout/AppShell.jsx
/documents	Documents	Event document		Authenticated route gate	src/App.jsx; src/layout/AppShell.jsx

Route	Title	Purpose	Component/Gate	Access	Source
		references, links, and evidence			
/contacts	Contacts & Organizations	Reusable people, businesses, and event relationships		Authenticated route gate	src/App.jsx; src/layout/AppShell.jsx
/event-review	Reports	Read-only follow-up, payments, operations, and summary		Authenticated route gate	src/App.jsx; src/layout/AppShell.jsx
/imports	Import Center	Bring in CSV exports and pasted table rows safely		Authenticated route gate	src/App.jsx; src/layout/AppShell.jsx
/qa	System QA	System health, data checks, and safe test guidance		Authenticated route gate	src/App.jsx; src/layout/AppShell.jsx
/communications	Message Builder	Create, personalize, and copy event messages		Authenticated route gate	src/App.jsx; src/layout/AppShell.jsx
/settings	Settings	Practical workspace and event defaults		Authenticated route gate	src/App.jsx; src/layout/AppShell.jsx



Caption: route map diagram.

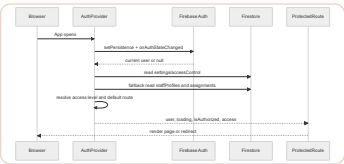
Public route: /login.

Redirect aliases: /security -> /settings, /reconciliation -> /payments/reconciliation, /reports -> /event-review.

Firebase Authentication

Firebase initialization is in `src/lib/firebase.js`. Authentication lifecycle is in `src/auth/AuthProvider.jsx`.

Auth Sequence



Caption: auth sequence diagram.

Definitions:

- Authentication state:
 Firebase's current determination of whether a user is signed in.
- Authorization state: Gathetr's determination that the signed-in user has owner, approved organizer, or assigned staff access.

Flow

- App initializes Firebase from Vite environment variables.
- Auth persistence is set to browser local persistence.
- Firebase reports the current user through `onAuthStateChanged`.
- Gathetr reads `settings/accessControl` for approved organizer access.
- If organizer access is not found, it reads `staffProfiles/{uid}` and active assignment docs for supported event IDs.
- `getUserAccessLevel` builds the access object used by route gates and labels.
- Firestore Rules still enforce every backend read/write.

Firestore Data Model

This dictionary is based on current services, Firestore Rules, and tests. It does not include real production personal data.

Collections and Subcollections

Collection path	Purpose	Code that uses it	Rules reference	Security note
settings/accessControl	Authoritative approved organizer source plus metadata records.	src/auth/AuthProvider.jsx; src/services/accessManagementService.js; src/pages/SettingsPage.jsx	firestore.rules match /settings/accessControl	Protected Owner only for mutations; approved admins read.
settings/accessControl/history/{historyId}	Append-only history for organizer access changes.	src/services/accessManagementService.js	firestore.rules match /settings/accessControl/history/{historyId}	Protected Owner create; approved admins read.
settings/integrations	Supported integration status settings surfaced in Settings.	src/services/integrationSettingsService.js	firestore.rules match /settings/integrations	Protected Owner create/update; approved admins read.
events/{eventId}	Event records, planning data, capability configuration, and working-event source.	src/services/eventService.js; src/events/ActiveEventProvider.jsx	firestore.rules match /events/{eventId}	Approved admins manage; assigned staff can read assigned events.
events/{eventId}/staffAssignments/{uid}	Event-scoped staff assignment records.	src/services/staffManagementService.js; src/auth/AuthProvider.jsx	firestore.rules match /events/{eventId}/staffAssignments/{uid}	Approved admins manage; assigned user can read own active assignment.
events/{eventId}/tasks/{taskId}	Event-scoped tasks and deadlines.	src/services/taskService.js	firestore.rules match /events/{eventId}/tasks/{taskId}	Approved admins and assigned event managers manage; viewers can read.
registrations/{registrationId}	Guest registration, ticket, payment, and check-in status records.	src/services/registrationService.js; src/services/ticketService.js	firestore.rules match /registrations/{registrationId}	Approved admins manage; assigned scanners can perform narrow check-in updates.
auditLogs/{logId}	Append-only audit evidence for business mutations.	src/services/auditService.js; service write batches	firestore.rules match /auditLogs/{logId}	Create only when matching a permitted target mutation; never update/delete.
operationsLedger/{ledgerEntryId}	Event-level Operations ledger, commitments, partners, in-kind support.	src/services/operationsLedgerService.js	firestore.rules match /operationsLedger/{ledgerEntryId}	Approved admins create/update; no delete.
events/{eventId}/documents/{documentId}	Event document references and external links. No Firebase Storage upload.	src/services/documentService.js	firestore.rules match /events/{eventId}/documents/{documentId}	Approved admins and event managers create/update; admins delete.
events/{eventId}/runOfShow/{itemId}	Event-day schedule sequence, dependencies, arrivals, status.	src/services/runOfShowService.js	firestore.rules match /events/{eventId}/runOfShow/{itemId}	Approved admins manage; assigned event staff can read through task read gate.
events/{eventId}/resources/{resourceId}	Equipment, supplies, packing, pickup,	src/services/eventResourceService.js	firestore.rules match /events/{eventId}/resources/{resourceId}	Approved admins manage; assigned

Collection path	Purpose	Code that uses it	Rules reference	Security note
	return, quantity tracking.			event staff can read through task read gate.
contacts/{contactId}	Reusable contact directory.	src/services/contactService.js	firestore.rules match /contacts/{contactId}	Approved admins create/update/read; delete false.
organizations/{organizationId}	Reusable organization directory.	src/services/contactService.js	firestore.rules match /organizations/{organizationId}	Approved admins create/update/read; delete false.
events/{eventId}/contactLinks/{linkId}	Event relationship links to contacts/organizations.	src/services/contactService.js	firestore.rules match /events/{eventId}/contactLinks/{linkId}	Approved admins and event managers create/update; admins delete.
accessRequests/{requestId}	Signed-in user access request workflow.	src/services/accessRequestContract.js	firestore.rules match /accessRequests/{requestId}	Signed-in create; approved admins review; no delete.
staffProfiles/{uid}	Global staff profile records.	src/services/staffManagementService.js; src/auth/AuthProvider.jsx	firestore.rules match /staffProfiles/{uid}	Approved admins manage; staff can read own active profile.

Indexes

Collection group	Fields	Purpose
registrations	eventId ASCENDING, createdAt DESCENDING	Supports scoped Firestore query ordering used by the app.

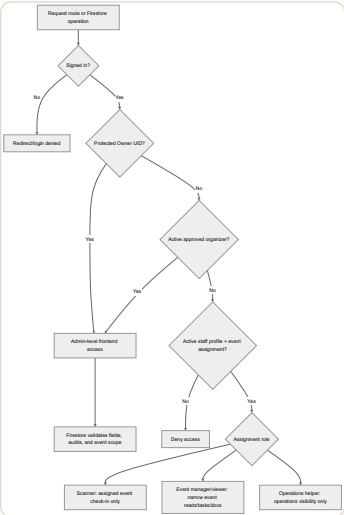
Schema Truth Rule

Do not infer schema from one example record. Cross-check service writes, page reads, utils, Firestore Rules validators, and tests.

Permissions and Security Rules

Permission architecture has two layers:

- 1. Frontend access checks in `src/utls/accessRoles.js` and `ProtectedRoute`.
- 2. Firestore Rules in `firestore.rules`.



Caption: permission flow diagram.

Permission Matrix

Action	Protected Owner	Approved Organizer/Admin	Assigned Staff	Scanner	Unapproved
Open organizer shell	Yes	Yes	Limited by assigned route	No, scanner uses /scanner	No
Manage Settings organizer access	Yes, immutable owner cannot be removed	No owner-only controls	No	No	No
Create/update events	Yes	Yes	No	No	No
Read assigned event	Yes	Yes	Yes if assignment active	Yes for scanner route	No
Create/update registrations	Yes	Yes	No	No except scanner check-in fields	No
Assign tickets	Yes	Yes	No	No	No
Complete check-in	Yes	Yes	No unless scanner assignment	Yes for assigned event	No

Action	Protected Owner	Approved Organizer/Admin	Assigned Staff	Scanner	Unapproved
Use Operations ledger	Yes	Yes	operations-helper read only	No	No
Use Import Center	Yes	Yes	No	No	No
Read documents/tasks as assigned staff	Yes	Yes	event-manager/viewer where allowed	No	No
Create audit logs	Only with matching mutation	Only with matching mutation	Only where rules allow matching mutation	Only check-in audit path	No

Firestore Rules Reference

Important helper functions include `isSignedIn`, `isProtectedOwner`, `isApprovedAdmin`, `activeStaffProfile`, `activeStaffAssignment`, `isAssignedScanner`, `canReadTask`, and `canManageTask`.

Rules distinguish `resource.data` from `request.resource.data`, and `create/read/update/delete` paths are intentionally different. Query rules are not filters: if a query can return forbidden documents, Firestore denies the whole query.

Events, Guests, Tickets, and Check-In

Event Management

Events are managed by `EventsPage` and `eventService`. The Working Event context determines which event-scoped pages load data.

Guest Registrations

Registrations include guest identity, persons attending, payment state, ticket status, historical attendance evidence, and scanner-confirmed check-in fields. Registration mutations must stay atomic with audit evidence.

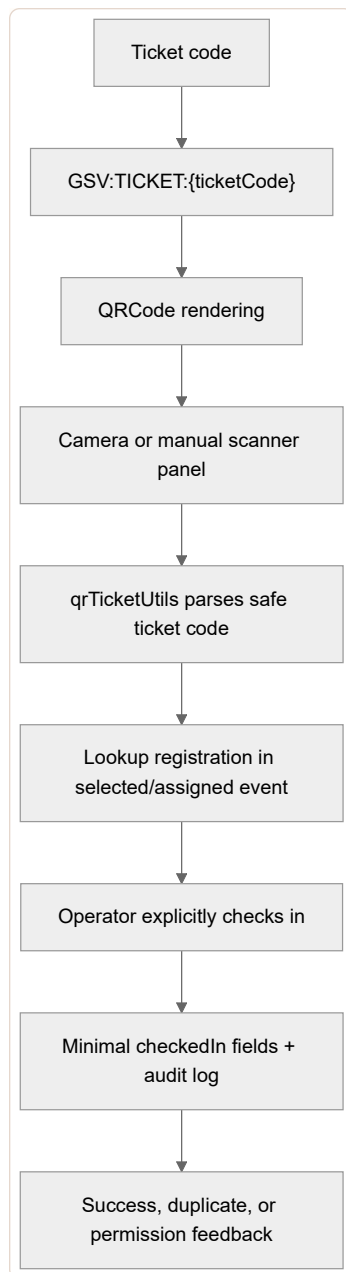
Tickets and QR

QR payload is exactly:

```
GSV:TICKET:{ticketCode}
```

Do not put private guest data into QR payloads.

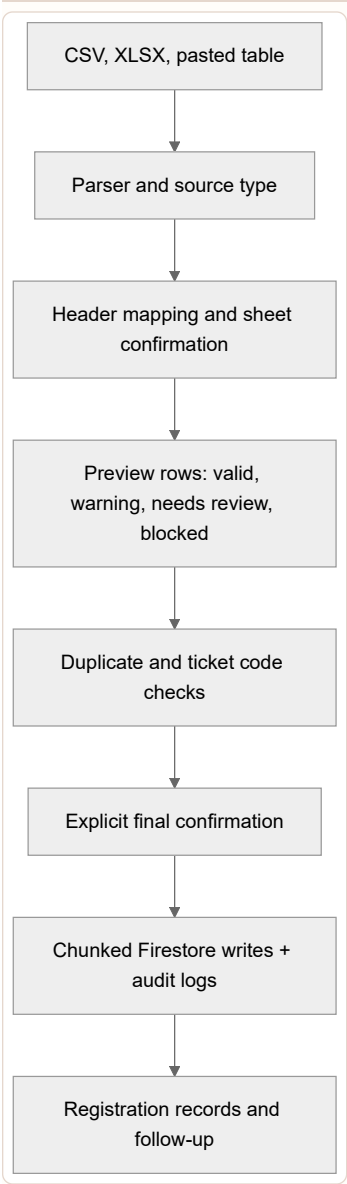
Check-In Flow



Caption: qr checkin flow diagram.

Imports, Exports, and QR Systems

Import Flow



Caption: import flow diagram.

Supported sources: CSV, XLSX through read-excel-file, and pasted table rows. The app uses preview-first validation, explicit sheet confirmation, duplicate detection, ticket-code collision checks, and chunked writes with audits.

Exports are client-side generated files from `src/utls/exportUtils.js` and related finance/reconciliation helpers. Audit log and access-control collections are not included in ordinary organizer exports.

Testing and Quality Assurance

Command Reference

Command	Purpose	Actual script	When to run	Expected result
npm run dev	Local validation/development command	vite	During development, QA, or documentation validation	Exit code 0; command-specific pass message
npm run build	Local validation/development command	vite build	During development, QA, or documentation validation	Exit code 0; command-specific pass message
npm run preview	Local validation/development command	vite preview	During development, QA, or documentation validation	Exit code 0; command-specific pass message
npm run docs:generate	Local validation/development command	node scripts/docs/generateTechnicalManual.mjs	During development, QA, or documentation validation	Exit code 0; command-specific pass message
npm run docs:validate	Local validation/development command	node scripts/docs/validateTechnicalDocs.mjs	During development, QA, or documentation validation	Exit code 0; command-specific pass message
npm run lint	Local validation/development command	eslint src tests scripts e2e eslint.config.js vite.config.js playwright.config.js	During development, QA, or documentation validation	Exit code 0; command-specific pass message
npm run test	Local validation/development command	node --test tests/*.test.js	During development, QA, or documentation validation	Exit code 0; command-specific pass message
npm run doctor	Local validation/development command	react-doctor . --project . --scope full --blocking none --no-telemetry	During development, QA, or documentation validation	Exit code 0; command-specific pass message
npm run doctor:changed	Local validation/development command	react-doctor . --project . --scope changed --base main --blocking error --no-telemetry	During development, QA, or documentation validation	Exit code 0; command-specific pass message

Command	Purpose	Actual script	When to run	Expected result
npm run doctor:json	Local validation/development command	react-doctor . --project . --scope full --blocking none --json --json-compact --no-telemetry	During development, QA, or documentation validation	Exit code 0; command-specific pass message
npm run product:copy-scan	Local validation/development command	node scripts/product/copyScan.mjs	During development, QA, or documentation validation	Exit code 0; command-specific pass message
npm run product:routes	Local validation/development command	node scripts/product/routeInventory.mjs	During development, QA, or documentation validation	Exit code 0; command-specific pass message
npm run product:bundle	Local validation/development command	node scripts/product/bundleSummary.mjs	During development, QA, or documentation validation	Exit code 0; command-specific pass message
npm run product:docs	Local validation/development command	node scripts/product/documentationCheck.mjs	During development, QA, or documentation validation	Exit code 0; command-specific pass message
npm run product:legacy	Local validation/development command	node scripts/product/legacyControlCheck.mjs	During development, QA, or documentation validation	Exit code 0; command-specific pass message
npm run product:qa	Local validation/development command	node scripts/product/runProductCommand.mjs qa	During development, QA, or documentation validation	Exit code 0; command-specific pass message
npm run product:audit	Local validation/development command	node scripts/product/runProductCommand.mjs audit	During development, QA, or documentation validation	Exit code 0; command-specific pass message
npm run product:qa:emulator-checks	Local validation/development command	node --test tests/firestore-checkin-rules.test.js tests/task-workflow-rules.test.js tests/document-contact-rules.test.js tests/run-of-show-resource-rules.test.js && npm run e2e:smoke:inner	During development, QA, or documentation validation	Exit code 0; command-specific pass message
npm run e2e:smoke	Local validation/development command	npx -y firebase-tools@14.19.0 emulators:exec --only auth,firestore --project gathervibeshub "npm run e2e:smoke:inner"	During development, QA, or documentation validation	Exit code 0; command-specific pass message
npm run e2e:smoke:inner	Local validation/development command	playwright test --project=chromium e2e/navigation.spec.js	During development, QA, or documentation validation	Exit code 0; command-specific pass message

Command	Purpose	Actual script	When to run	Expected result
npm run e2e:full	Local validation/development command	npx -y firebase-tools@14.19.0 emulators:exec --only auth,firestore --project gathervibeshub "playwright test --project=chromium"	During development, QA, or documentation validation	Exit code 0; command-specific pass message
npm run e2e:firefox	Local validation/development command	npx -y firebase-tools@14.19.0 emulators:exec --only auth,firestore --project gathervibeshub "playwright test --project=firefox"	During development, QA, or documentation validation	Exit code 0; command-specific pass message
npm run e2e:webkit	Local validation/development command	npx -y firebase-tools@14.19.0 emulators:exec --only auth,firestore --project gathervibeshub "playwright test --project=webkit"	During development, QA, or documentation validation	Exit code 0; command-specific pass message
npm run admin:ensure-access	Local validation/development command	node scripts/admin/ensureAccessControl.mjs	During development, QA, or documentation validation	Exit code 0; command-specific pass message
npm run admin:verify-firebase	Local validation/development command	node scripts/admin/verifyFirebaseSetup.mjs	During development, QA, or documentation validation	Exit code 0; command-specific pass message
npm run admin:replace-codex-test-with-demo	Local validation/development command	node scripts/admin/replaceCodexTestWithDemoEvent.mjs	During development, QA, or documentation validation	Exit code 0; command-specific pass message
npm run admin:verify-production-fixtures	Local validation/development command	node scripts/admin/verifyProductionFixtures.mjs	During development, QA, or documentation validation	Exit code 0; command-specific pass message
npm run admin:verify-production-counts	Local validation/development command	node scripts/admin/verifyProductionCounts.mjs	During development, QA, or documentation validation	Exit code 0; command-specific pass message
npm run firebase:deploy-rules	PRODUCTION-IMPACTING COMMAND where noted	npx firebase-tools deploy --only firestore:rules,firestore:indexes --project gathervibeshub	Only after explicit approval and validation	Exit code 0; command-specific pass message
npm run firebase:deploy-hosting	PRODUCTION-IMPACTING COMMAND where noted	npx firebase-tools deploy --only hosting --project gathervibeshub	Only after explicit approval and validation	Exit code 0; command-specific pass message
npm run firebase:deploy-all	PRODUCTION-IMPACTING COMMAND where noted	npx firebase-tools deploy --only firestore:rules,firestore:indexes,hosting --project gathervibeshub	Only after explicit approval and validation	Exit code 0; command-specific pass message

Testing Layers

- Unit and source contract tests: `tests/*.test.js`.
- Firestore Rules tests: rules-specific test files run against emulators.
- E2E smoke: `e2e/navigation.spec.js`.
- Accessibility and responsive E2E: `e2e/accessibility.spec.js`, `e2e/responsive.spec.js`.
- Product QA wrapper: `npm run product:qa`.
- React Doctor: `npm run doctor:json` or `npm run doctor:changed`.

Date-Sensitive Test Policy

Phase 2 found `tests/document-contact-foundation.test.js` used `2026-08-20` as an expiring-soon fixture. On `2026-08-21` that fixture became expired for helpers using the current clock. It was updated to `2026-08-30` to preserve the intended behavior.

Future date-sensitive tests should either inject an explicit clock into every helper under test or choose dates far enough away to avoid silent expiry.

Build, Deployment, and Recovery

Local Build

```
npm ci
npm run lint
npm test
npm run build
npm run product:qa
```

Firebase Project

Default project in .firebaserc:
gathervibeshub.

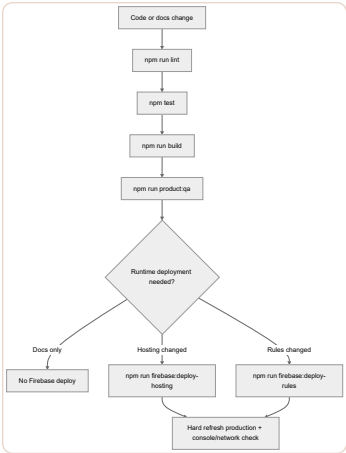
Hosting public folder: dist.

Security headers and SPA rewrite are
configured in firebase.json.

Production-Impacting Commands

Command	Impact	Prerequisites
npm run firebase:deploy- hosting	PRODUCTION- IMPACTING COMMAND: deploys Hosting assets from dist.	Build passed; production visual check plan ready.
npm run firebase:deploy- rules	PRODUCTION- IMPACTING COMMAND: deploys Firestore Rules and indexes.	Rules tests passed; authorization impact reviewed.
npm run firebase:deploy- all	PRODUCTION- IMPACTING COMMAND: deploys Hosting, Rules, and indexes.	Use only when all targets intentionally changed.

Deployment Flow



Caption: deployment flow diagram.

No production deployment is performed by documentation generation.

Troubleshooting and Repairs

Use the runbooks in `docs/runbooks/` for step-by-step procedures.

Priority problems:

- Blank page or stale deployment chunk.
- Login or auth persistence failure.
- Missing or insufficient permissions.
- Owner account denied.
- Staff assignment missing.
- Scanner cannot check in.
- Firestore permits data that frontend hides or frontend shows data Firestore denies.
- Build, dependency, emulator, or deployment failure.
- QR scanner failure during an event.

Legacy Systems and Technical Debt

Current Register

Item	Classification	Notes
Archived migration evidence	archive only / future review required	Located at C:\Users\Jaylan\Documents\Archived Projects\gsv-codex-demo-migration-evidence-2026-08-21. Contains CPB/CODEX_DEMO evidence and old repo snapshots.
Legacy repository archive	archive only	Located at C:\Users\Jaylan\Documents\Archived Projects\gathervibes-legacy-repository-archive-2026-08-21.
21 node_modules in historical archive snapshots	future review required	Generated dependencies likely reclaimable, but recursive deletion was blocked in Phase 2.
Historical branches	future review required	Branches were not deleted in Phase 2. Review merge status before any cleanup.
Active copilot worktree	current / unknown future work	Retained at C:\Users\Jaylan\Documents\gathetr.worktrees\copilot-personalized-welcome-tour-implementation.
React Doctor warnings	technical debt	doctor.json reports warnings, but doctor:changed found no issues for Phase 2 changed file.
Legacy role aliases	legacy but required	accessRoles retains aliases such as checkInStaff for compatibility.

Change Management

No meaningful coding task is complete until documentation impact has been reviewed.

Every handoff must report:

- Documentation reviewed:
YES/NO
- Documentation changed:
YES/NO
- Documentation files changed
- Reason no documentation
update was required

Use templates in docs/templates/ for change records, incidents, repair runbooks, release checklists, and manual update checklists.

Reference and Glossary

Glossary

- Approved Organizer: an active email in settings/accessControl.
- Protected Owner: the permanent UID-protected owner account, independent of mutable allowlists.
- Working Event: selected event context used by event-scoped pages.
- CODEX_DEMO: synthetic training event safe for reversible QA.
- CPB: real completed event; not a synthetic QA target.
- Audit log: append-only evidence document required for many business writes.
- Query rules are not filters: Firestore denies a query if it could return unauthorized documents.

Search Keywords

Firebase missing or insufficient permissions resource.data request.resource.data
Firestore query rules are not filters
React Firebase auth state loading before Firestore query
Vite dynamic import Failed to fetch dynamically imported module
Firebase Hosting stale service worker chunk cache
Playwright Firebase emulator auth firestore port taken 9099 8080
html5-qrcode camera permission HTTPS mobile browser
GSV:TICKET QR payload ticketCode privacy
read-excel-file browser XLSX import mapping
React Doctor no-loading-flag-reset-outside-finally

Future App Development Reference

Lessons from Gathetr that apply to future React/Firebase apps:

- Design the permission model before building pages.
- Keep frontend route gates and backend Rules aligned but separate.
- Use one authoritative source for access control and display it in Settings.
- Avoid synthetic QA against real event data.
- Keep QR payloads privacy-safe and stable.
- Add emulator-backed rule tests before deploying Rules.
- Document every schema, role, status, and workflow change at the time it changes.
- Keep generated PDFs as snapshots, not the source of truth.

Runbook 1.

Application and Session Failures

Purpose

Repair organizer-facing failures where the app does not boot correctly, loses session state, or loops back to login.

Symptoms

- Blank page after navigation.
- Failed to fetch dynamically imported module.
- Login succeeds briefly, then the app returns to /login.
- Session does not persist across refresh.
- Firebase appears unavailable in the browser.

Severity

High.

Possible causes

- Stale browser cache after a Hosting release.
- Broken dynamically imported bundle.
- Firebase config mismatch.
- Persistence setup failure in AuthProvider.
- Browser storage or cookie restrictions.

Safety warnings

- Do not deploy while diagnosing unless the fix is already proven.
- Do not clear production data or mutate auth state to test a theory.

Evidence to collect

- Current route and query string.
- Console error text.
- Network failures for JS chunks or Firebase requests.

- Signed-in user state if visible.
- App build commit shown by diagnostics where available.

First checks

1. Confirm whether the issue reproduces after hard refresh.
2. Check console for dynamic-import, Firebase, or storage errors.
3. Check whether the browser is offline or unstable.
4. Confirm the route is expected for the current role and Working Event.

Files to inspect

- `src/components/AppErrorBoundary.jsx`
- `src/utils/appErrorDiagnostics.js`
- `src/lib/firebase.js`
- `src/auth/AuthProvider.jsx`
- `src/auth/ProtectedRoute.jsx`
- `firebase.json`

Commands to run

- `npm run build`
- `npm run product:qa`
- `npm run e2e:smoke`
- `npm run admin:verify-firebase` only if a safe production-read check is justified

Step-by-step diagnosis

1. If the screen is blank and console shows a failed chunk fetch, treat it as stale deployment drift first.
2. Reload with a cache-busting navigation and confirm the requested JS asset exists in the current build output.
3. If the app returns to `/login`, inspect `AuthProvider` persistence setup and the computed return route.
4. If Firebase initialization fails, verify `VITE_FIREBASE_*` values are present in the active environment without exposing secrets.
5. If the browser loses auth only on refresh, inspect local

persistence restrictions or
browser storage failures.

Repair options

- Stale chunk: redeploy only the correct hosting build in a later release phase, then hard-refresh.
- Session failure: correct persistence or return-path logic and retest locally.
- Firebase boot issue: correct local environment or document missing configuration.

Verification

- Organizer can load /dashboard after refresh without losing session.
- No dynamic-import failure remains in console.
- `npm run product:qa` and `npm run e2e:smoke` pass.

Rollback

- Revert the bad frontend commit or restore the previous Hosting release.

Escalation conditions

- Protected Owner cannot sign in.
- Dynamic-import failure persists across clean local rebuild and validated release artifact.
- Firebase configuration appears broken in production.

Search keywords

- blank page
- dynamic import
- stale deployment chunk
- auth persistence
- session lost
- return to login

Related tests

- `tests/error-boundary-classification*.test.js`

- `tests/auth-reliability.test.js`
- `e2e/navigation.spec.js`

Related manual sections

- Application Overview
- Firebase Authentication
- Build, Deployment, and Recovery

Runbook 2.

Permission and Access Failures

Purpose

Repair access failures involving Protected Owner, approved organizers, staff profiles, staff assignments, and rule/frontend mismatches.

Symptoms

- Protected Owner denied in Settings or core organizer routes.
- Approved organizer denied despite visible record in Settings.
- Staff profile not recognized.
- Staff event assignment missing or ignored.
- Frontend allows a route but Firestore denies access.
- Firestore allows access that the frontend hides.
- Query denied because Firestore Rules are not filters.

Severity

High.

Possible causes

- settings/accessControl record mismatch.
- Organizer status not active.
- UID-based Protected Owner bypass missing from app state or misunderstood in diagnosis.
- Inactive staffProfiles/{uid}.
- Missing or inactive events/{eventId}/staffAssignments/{uid}.
- Query shape too broad for rule visibility.
- Route gate logic drift from Rules.

Safety warnings

- Never weaken Rules to make a query succeed.

- Never remove or mutate the Protected Owner account as a debugging shortcut.

Evidence to collect

- Signed-in email and UID.
- Access level resolved in the app.
- Target route and target collection/query.
- Exact Firestore permission error text.
- Relevant Settings screen status.

First checks

1. Confirm whether the user is Protected Owner, approved organizer, or staff.
2. Confirm organizer status is active in `settings/accessControl`.
3. Confirm staff profile status is active.
4. Confirm event assignment exists and matches the target event.

Files to inspect

- `src/auth/AuthProvider.jsx`
- `src/utils/accessRoles.js`
- `src/auth/ProtectedRoute.jsx`
- `src/services/accessManagementService.js`
- `src/services/staffManagementService.js`
- `firestore.rules`

Commands to run

- `npm test`
- `npm run product:qa`
- `npm run admin:verify-production-fixtures` only for safe production reads

Step-by-step diagnosis

1. For Protected Owner denial, confirm the signed-in UID matches the immutable owner UID and that the app state still marks owner capability `true`.
2. For approved organizer denial, inspect `approvedEmails`,

`rolesByEmail`, and `approvedOrganizerRecords` together; only active records should authorize.

3. For staff denial, confirm both the global profile and event assignment are active and use the same UID/event id pair.
4. If the frontend shows the route but Firestore denies, compare the query or write shape against the relevant rule block and reduce scope rather than broadening Rules.
5. If Firestore would allow but the UI hides, compare `canViewRoute` capability mapping with the Rules-backed intent and update the app contract, not live data.

Repair options

- Correct stale organizer/staff metadata in approved owner-only settings tooling.
- Correct query scope so Firestore Rules can evaluate it safely.
- Update route/capability logic to match existing Rules when the backend contract is already correct.
- Document unsupported legacy records as UNKNOWN — REQUIRES FUTURE VERIFICATION if proof is missing.

Verification

- Target user reaches only the routes their role should support.
- Matching Firestore reads/writes succeed or fail exactly as intended.
- Settings still prevents disabling or removing Protected Owner.

Rollback

- Revert the route/access logic change or restore prior approved organizer metadata.

Escalation conditions

- Protected Owner cannot recover access.
- A fix would require broadening Rules without narrow proof.
- The same query must read mixed-visibility documents.

Search keywords

- missing or insufficient permissions
- protected owner denied
- approved organizer denied
- staff assignment missing
- rules are not filters

Related tests

- tests/protected-owner-authorization-matrix.test.js
- tests/settings-access-management.test.js
- Rules tests covering registrations, tasks, documents, contacts, run-of-show, and resources

Related manual sections

- Firebase Authentication
- Permissions and Security Rules
- Settings and staff access guidance

Runbook 3.

Scanner, Ticket, and Check-In Failures

Purpose

Repair event-day failures involving scanner access, camera use, QR parsing, ticket lookup, and duplicate attendance handling.

Symptoms

- Scanner route opens but no assigned event loads.
- Browser camera cannot open.
- QR code scans but is not recognized.
- Ticket lookup fails.
- Duplicate check-in warning appears.

Severity

High during live operations.

Possible causes

- Missing active scanner assignment.
- Browser camera permission denied or insecure origin.
- QR payload not using `GSV:TICKET:{ticketCode}`.
- Ticket code missing, invalid, or duplicated.
- Registration already checked in.

Safety warnings

- Do not change QR format during live diagnosis.
- Do not use CPB or real production attendees for synthetic scanner rehearsal.

Evidence to collect

- Event id and scanner user.
- Camera permission state.
- Scanned payload text if safely visible.

- Registration `ticketCode`, `ticketStatus`, `checkedIn`, and `checkInTime`.

First checks

1. Confirm the scanner user has an active assignment for the target event.
2. Confirm the page is running on HTTPS or a trusted localhost context.
3. Confirm the scanned value starts with `GSV:TICKET:.`
4. Confirm the registration exists in the selected or assigned event.

Files to inspect

- `src/pages/ScannerPage.jsx`
- `src/pages/CheckInPage.jsx`
- `src/components/checkin/QrScannerPanel.jsx`
- `src/utils/qrTicketUtils.js`
- `src/utils/checkInUtils.js`
- `src/services/ticketService.js`

Commands to run

- `npm test -- tests/phase14-camera-checkin.test.js`
- `npm test -- tests/phase7-qr-checkin.test.js`
- `npm run e2e:smoke`

Step-by-step diagnosis

1. For missing assigned event, inspect resolved assignments in app state and confirm event id equality.
2. For camera failure, inspect browser permission state and HTTPS context before touching app code.
3. For unrecognized QR, parse the value with `qrTicketUtils` and compare it to the registration list for the active event only.
4. For lookup failure, confirm the registration has a current `ticketCode` and that duplicate ticket codes were not imported.

5. For duplicate attendance, inspect `checkedIn`, `checkInTime`, and audit history before offering undo.

Repair options

- Correct assignment/profile state.
- Correct QR payload generation or lookup code while preserving the prefix standard.
- Regenerate or remove a bad ticket assignment through the ticket workflow.
- Use the narrow undo check-in path when attendance was recorded in error.

Verification

- Scanner opens the assigned event correctly.
- Camera path can scan a valid QR.
- Check-in writes only the approved narrow attendance fields plus audit log.

Rollback

- Restore the previous ticket code state or revert the event-day UI change.

Escalation conditions

- QR payload format appears to have drifted.
- Duplicate ticket codes exist across active event registrations.
- Check-in writes require broader Rules changes.

Search keywords

- scanner cannot open assigned event
- camera permission denied
- QR not recognized
- ticket lookup fails
- duplicate check-in

Related tests

- `tests/phase14-camera-checkin.test.js`

- `tests/phase7-qr-checkin.test.js`
- `tests/firestore-checkin-rules.test.js`

Related manual sections

- Events, Guests, Tickets, and Check-In
- Imports, Exports, and QR Systems

Runbook 4. Import and Data Repair

Purpose

Repair import failures, invalid mapping, duplicate registration detection, ticket collisions, and legacy record shape issues.

Symptoms

- Registration import fails before confirm or during chunked write.
- Spreadsheet mapping is invalid.
- Duplicate registration import warnings block progress.
- Ticket-code collision occurs.
- Old Firestore record is missing ownership or permission fields.

Severity

Medium to High depending on whether live registration intake is blocked.

Possible causes

- Unsupported sheet/header shape.
- Missing event price or finance data.
- Duplicate sourceRowId, ticket code, or strong contact match.
- Legacy records missing modern optional fields.
- Import preview data not normalized before write.

Safety warnings

- Never bypass preview and validation to force-import data.
- Never bulk-write to production while the duplicate or ticket-collision root cause is unknown.

Evidence to collect

- Source type and selected sheet.

- Header mapping status.
- Preview row classifications.
- Exact blocking row ids and ticket codes.
- Existing registration ids involved in collisions.

First checks

1. Confirm the selected source type matches the file or pasted data.
2. Confirm the sheet selection step completed for XLSX sources.
3. Confirm duplicate warnings are event-scoped, not cross-event assumptions.
4. Confirm ticket codes are unique within the active event.

Files to inspect

- `src/pages/ImportsPage.jsx`
- `src/services/importService.js`
- `src/utils/importUtils.js`
- `src/utils/xlsxImport.js`
- `src/utils/validators.js`
- `src/utils/qrTicketUtils.js`

Commands to run

- `npm test -- tests/import-center.test.js`
- `npm test -- tests/import-center-workflow-upgrade.test.js`
- `npm test -- tests/xlsx*.test.js`

Step-by-step diagnosis

1. Reproduce with a local copy of the source file and confirm where the workflow stops: source, mapping, validation, review, confirm, or result.
2. Compare incoming headers to the source-type expectations already defined in the app.
3. Inspect duplicate classification: soft warning, needs review, or hard block.

4. For ticket collisions, compare imported ticket codes to existing event registrations and earlier rows in the same batch.
5. For old records, normalize on edit rather than mutating historical production data in bulk without approval.

Repair options

- Add or correct header mapping support.
- Improve preview normalization or safe defaults.
- Fix duplicate logic if it over-classifies across unrelated event rows.
- Remove or regenerate conflicting ticket codes in the proper ticket flow.

Verification

- Import preview reaches final confirm without hidden writes.
- Confirmed import writes only valid event-scoped registration rows.
- Duplicate and ticket-collision protections remain active.

Rollback

- Revert the import parser or mapping change and preserve the failing sample for future repair.

Escalation conditions

- A fix would require rewriting historical production records without review.
- The imported source contains private or inconsistent data that cannot be normalized safely.

Search keywords

- import failure
- spreadsheet mapping invalid
- duplicate registration
- ticket collision
- legacy record missing fields

Related tests

- Import Center test suite
- finance and QR privacy tests

Related manual sections

- Imports, Exports, and QR Systems
- Field-Level Firestore Dictionary

Runbook 5.

Development, Build, and Test Failures

Purpose

Repair local dependency, build, emulator, unit-test, Firestore Rules test, Playwright, and React Doctor failures.

Symptoms

- Dependency install fails.
- Build fails.
- Emulator port collision.
- Unit tests fail.
- Firestore Rules tests fail.
- Playwright/E2E fails.
- React Doctor warning requires review.

Severity

Medium for local-only work, High when blocking release validation.

Possible causes

- Node dependency mismatch.
- Vite import failure or unresolved module.
- Java or emulator process conflict.
- Stale fixture assumptions.
- Rule/test contract drift.
- Local browser runner instability.

Safety warnings

- Do not treat an emulator-only failure as proof of production failure without corroboration.
- Do not ignore React Doctor blocking findings on changed code.

Evidence to collect

- Exact failing command.
- Exit code and first actionable stack line.
- Occupied ports.

- Last passing commit if known.

First checks

1. Confirm clean install state with current lockfile.
2. Confirm ports 8080 and 9099 are free before emulator runs.
3. Confirm the failing test is not using an expired date-sensitive fixture.

Files to inspect

- `package.json`
- `package-lock.json`
- `vite.config.js`
- `playwright.config.js`
- `firestore.rules`
- The failing test file and nearest helper

Commands to run

- `npm run lint`
- `npm test`
- `npm run build`
- `npm run product:qa`
- `npm run doctor:changed`

Step-by-step diagnosis

1. Re-run the narrow failing command first.
2. If the failure is emulator startup, stop the stale process rather than changing tests immediately.
3. If the failure is build-time import resolution, inspect the exact file path and package classification.
4. If the failure is rules-related, compare the intended write shape against the rule validator and existing rule tests.
5. If Playwright fails, isolate whether the problem is app state, test selector drift, or browser infrastructure.

Repair options

- Correct dependency classification or lockfile.

- Fix date-sensitive test fixtures.
- Repair route selector or app contract drift.
- Adjust local QA wrapper only if it improves determinism without weakening checks.

Verification

- Failing narrow command passes.
- Full repo validation set passes for the scoped change.

Rollback

- Revert the failing tooling or test change, then restore the previous passing workflow.

Escalation conditions

- The same failure reproduces across clean installs and multiple recent commits.
- QA tooling requires unsafe system changes.

Search keywords

- dependency install failure
- build failed
- emulator port 8080 9099
- unit test failed
- Firestore Rules test failed
- Playwright failed
- React Doctor warning

Related tests

- Entire project test and QA suite

Related manual sections

- Testing and Quality Assurance
- Build, Deployment, and Recovery

Runbook 6.

Deployment, Firebase, and Project Targeting

Purpose

Repair problems caused by the wrong Firebase project, invalid environment configuration, production Hosting regression, or incorrect Rules deployment.

Symptoms

- Wrong Firebase project selected.
- Production Hosting regression.
- Missing or invalid environment configuration.
- Incorrect Firestore Rules deployment.

Severity

High.

Possible causes

- Wrong gathervibeshub project selection.
- Build artifact mismatch.
- Missing VITE_FIREBASE_* values locally.
- Rules deployed without matching tests or diff review.

Safety warnings

- Never deploy during diagnosis without explicit scope approval.
- Never use a broad deploy when only one target changed.

Evidence to collect

- Current branch and commit.
- Active Firebase project.
- Build artifact version/commit.
- Exact production symptom and console/network evidence.

First checks

1. Confirm `.firebaserc` target and deploy command arguments.
2. Confirm whether the issue is Hosting-only, Rules-only, or both.
3. Confirm the current build passes locally before any deployment thinking.

Files to inspect

- `.firebaserc`
- `firebase.json`
- `src/lib/firebase.js`
- `firestore.rules`
- `dist/`

Commands to run

- `npm run build`
- `npm run product:qa`
- `npm run admin:verify-firebase`
- Do not run `firebase:deploy-*` unless explicitly approved

Step-by-step diagnosis

1. Confirm the project id is `gathervibeshub` and that the symptom is not local-environment-only.
2. If Hosting regressed, compare the built asset names and current app chunk requests.
3. If Rules regressed, inspect the changed rule block and the matching emulator tests.
4. If environment values are missing locally, fix the local configuration rather than changing production behavior.

Repair options

- Correct local project targeting.
- Rebuild and verify the exact artifact intended for Hosting.
- Revert or narrow the Rules change in a later approved phase.

Verification

- Local build and QA pass.
- Safe production-read checks align with expected Firebase target.

Rollback

- Restore the previous validated Hosting release or Rules version.

Escalation conditions

- Production behavior is broken but local source cannot reproduce.
- A deploy is requested without clear target isolation.

Search keywords

- wrong Firebase project
- hosting regression
- invalid environment config
- incorrect Rules deployment

Related tests

- `npm run product:qa`
- relevant emulator Rules suites

Related manual sections

- [Firebase Authentication](#)
- [Build, Deployment, and Recovery](#)

Runbook 7. Git Worktree and Local Recovery

Purpose

Repair local repository problems including worktree confusion and apparent loss of uncommitted work.

Symptoms

- Git worktree confusion.
- Lost or uncommitted local changes.

Severity

Medium to High depending on whether unrecoverable local work is at risk.

Possible causes

- Editing the wrong worktree.
- Detached expectations about main versus a docs branch.
- Untracked generated artifacts masking real diffs.

Safety warnings

- Do not use `git reset --hard` as a first response.
- Do not delete worktrees or untracked files until the owner confirms they are disposable.

Evidence to collect

- `git status --short --branch`
- `git worktree list --porcelain`
- `git rev-parse HEAD`
- Paths of modified or untracked files

First checks

1. Confirm the current repository path.
2. Confirm the current branch name.
3. Confirm whether the missing work was ever committed in another worktree.

Files to inspect

- `.git`
- Worktree list output
- Any changed source or artifact folders involved in the confusion

Commands to run

- `git status --short --branch`
- `git worktree list --porcelain`
- `git log --oneline --decorate -n 20`

Step-by-step diagnosis

1. Identify the active worktree and its branch.
2. Check whether the missing change exists in another local worktree or recent commit.
3. Separate generated artifacts from handwritten source changes.
4. Stage or copy only after the real source of truth is identified.

Repair options

- Switch to the intended branch/worktree.
- Commit recoverable work on a safe branch.
- Remove only confirmed disposable generated artifacts.

Verification

- The correct branch/worktree contains the intended source changes.
- `git status` is understandable and no owner work was destroyed.

Rollback

- Restore from the branch or commit where the work last existed.

Escalation conditions

- A required worktree is missing or corrupted.
- The only apparent recovery path would discard unreviewed owner changes.

Search keywords

- git worktree confusion
- lost local changes
- wrong branch
- clean tree mismatch

Related tests

- No direct tests; use Git evidence and repo validation after recovery.

Related manual sections

- Change Management
- Documentation Maintenance Procedure

Runbook 8.

Documentation Generation and PDF Repair

Purpose

Repair failures in the technical-manual pipeline itself, including HTML rendering, diagram rendering, bookmark generation, and PDF quality validation.

Symptoms

- Documentation generation fails.
- Raw Markdown appears in HTML or PDF.
- Mermaid appears as source text.
- PDF bookmarks are missing.
- TOC or internal links are broken.

Severity

Medium for local docs work, High when the owner manual is the intended release artifact.

Possible causes

- Missing docs dependencies.
- Markdown parser regression.
- Mermaid render failure.
- Bookmark post-processing failure.
- PDF helper dependency missing.

Safety warnings

- Do not patch the generated PDF by hand.
- Fix the generator or source docs, then regenerate.

Evidence to collect

- Failing docs command.
- Generated HTML excerpt showing bad output.
- PDF inspection metrics from `docs:pdf-check`.

- Rendered page PNGs and contact sheet.

First checks

1. Run `npm run docs:generate`.
2. Run `npm run docs:validate`.
3. Run `npm run docs:pdf-check`.
4. Inspect the generated HTML and first few PDF pages.

Files to inspect

- `scripts/docs/generateTechnicalManual.mjs`
- `scripts/docs/validateTechnicalDocs.mjs`
- `scripts/docs/pdfCheck.mjs`
- `scripts/docs/pdfTools.py`
- `docs/manual/*`
- `docs/runbooks/*`
- `docs/appendices/*`

Commands to run

- `npm run docs:generate`
- `npm run docs:validate`
- `npm run docs:pdf-check`

Step-by-step diagnosis

1. Confirm the source Markdown itself is correct and free of accidental raw HTML comments.
2. Confirm the Markdown parser is creating real table and heading HTML.
3. Confirm each Mermaid source file rendered to a real SVG before PDF generation.
4. Confirm the PDF helper can inspect text, add bookmarks, and render page PNGs.
5. Confirm the final PDF has no near-empty accidental pages.

Repair options

- Restore or reinstall missing docs dependencies.
- Fix the Markdown/diagram transformation step.
- Update appendix or runbook source content if the

generator is behaving correctly.

Verification

- `docs:generate`, `docs:validate`, and `docs:pdf-check` all pass.
- First page is a real cover and no raw source markers remain.

Rollback

- Revert the generator or source-doc commit that introduced the rendering regression.

Escalation conditions

- PDF bookmarks cannot be generated with the selected local pipeline.
- Required screenshots or diagrams cannot be rendered readably after reasonable correction.

Search keywords

- documentation generation failure
- raw Markdown in PDF
- Mermaid not rendered
- bookmark missing
- PDF quality validation failed

Related tests

- `npm run docs:validate`
- `npm run docs:pdf-check`

Related manual sections

- Documentation Maintenance Procedure
- Build, Deployment, and Recovery

Firestore Data Dictionary

This dictionary is based on current services, Firestore Rules, and tests. It does not include real production personal data.

Collections and Subcollections

Collection path	Purpose	Code that uses it	Rules reference	Security note
settings/accessControl	Authoritative approved organizer source plus metadata records.	src/auth/AuthProvider.jsx; src/services/accessManagementService.js; src/pages/SettingsPage.jsx	firestore.rules match /settings/accessControl	Protected Owner only for mutations; approved admins read.
settings/accessControl/history/{historyId}	Append-only history for organizer access changes.	src/services/accessManagementService.js	firestore.rules match /settings/accessControl/history/{historyId}	Protected Owner create; approved admins read.
settings/integrations	Supported integration status settings surfaced in Settings.	src/services/integrationSettingsService.js	firestore.rules match /settings/integrations	Protected Owner create/update; approved admins read.
events/{eventId}	Event records, planning data, capability configuration, and working-event source.	src/services/eventService.js; src/events/ActiveEventProvider.jsx	firestore.rules match /events/{eventId}	Approved admins manage; assigned staff can read assigned events.
events/{eventId}/staffAssignments/{uid}	Event-scoped staff assignment records.	src/services/staffManagementService.js; src/auth/AuthProvider.jsx	firestore.rules match /events/{eventId}/staffAssignments/{uid}	Approved admins manage; assigned user can read own active assignment.
events/{eventId}/tasks/{taskId}	Event-scoped tasks and deadlines.	src/services/taskService.js	firestore.rules match /events/{eventId}/tasks/{taskId}	Approved admins and assigned event managers manage; viewers can read.
registrations/{registrationId}	Guest registration, ticket, payment, and check-in status records.	src/services/registrationService.js; src/services/ticketService.js	firestore.rules match /registrations/{registrationId}	Approved admins manage; assigned scanners can perform narrow check-in updates.
auditLogs/{logId}	Append-only audit evidence for business mutations.	src/services/auditService.js; service write batches	firestore.rules match /auditLogs/{logId}	Create only when matching a permitted target mutation; never update/delete.
operationsLedger/{ledgerEntryId}	Event-level Operations ledger, commitments, partners, in-kind support.	src/services/operationsLedgerService.js	firestore.rules match /operationsLedger/{ledgerEntryId}	Approved admins create/update; no delete.
events/{eventId}/documents/{documentId}	Event document references and external links. No Firebase Storage upload.	src/services/documentService.js	firestore.rules match /events/{eventId}/documents/{documentId}	Approved admins and event managers create/update; admins delete.
events/{eventId}/runOfShow/{itemId}	Event-day schedule sequence, dependencies, arrivals, status.	src/services/runOfShowService.js	firestore.rules match /events/{eventId}/runOfShow/{itemId}	Approved admins manage; assigned event staff can read through task read gate.
events/{eventId}/resources/{resourceId}	Equipment, supplies, packing, pickup,	src/services/eventResourceService.js	firestore.rules match /events/{eventId}/resources/{resourceId}	Approved admins manage; assigned

Collection path	Purpose	Code that uses it	Rules reference	Security note
	return, quantity tracking.			event staff can read through task read gate.
contacts/{contactId}	Reusable contact directory.	src/services/contactService.js	firestore.rules match /contacts/{contactId}	Approved admins create/update/read; delete false.
organizations/{organizationId}	Reusable organization directory.	src/services/contactService.js	firestore.rules match /organizations/{organizationId}	Approved admins create/update/read; delete false.
events/{eventId}/contactLinks/{linkId}	Event relationship links to contacts/organizations.	src/services/contactService.js	firestore.rules match /events/{eventId}/contactLinks/{linkId}	Approved admins and event managers create/update; admins delete.
accessRequests/{requestId}	Signed-in user access request workflow.	src/services/accessRequestContract.js	firestore.rules match /accessRequests/{requestId}	Signed-in create; approved admins review; no delete.
staffProfiles/{uid}	Global staff profile records.	src/services/staffManagementService.js; src/auth/AuthProvider.jsx	firestore.rules match /staffProfiles/{uid}	Approved admins manage; staff can read own active profile.

Indexes

Collection group	Fields	Purpose
registrations	eventId ASCENDING, createdAt DESCENDING	Supports scoped Firestore query ordering used by the app.

Schema Truth Rule

Do not infer schema from one example record. Cross-check service writes, page reads, utils, Firestore Rules validators, and tests.

Permission architecture has two layers:

1. Frontend access checks in `src/utils/accessRoles.js` and `ProtectedRoute`.
2. Firestore Rules in `firestore.rules`.



Action	Protected Owner	Approved Organizer/Admin	Assigned Staff	Scanner	Unapproved
Open organizer shell	Yes	Yes	Limited by assigned route	No, scanner uses /scanner	No
Manage Settings organizer access	Yes, immutable owner cannot be removed	No owner-only controls	No	No	No
Create/update events	Yes	Yes	No	No	No
Read assigned event	Yes	Yes	Yes if assignment active	Yes for scanner route	No
Create/update registrations	Yes	Yes	No	No except scanner check-in fields	No
Assign tickets	Yes	Yes	No	No	No
Complete check-in	Yes	Yes	No unless scanner assignment	Yes for assigned event	No

Action	Protected Owner	Approved Organizer/Admin	Assigned Staff	Scanner	Unapproved
Use Operations ledger	Yes	Yes	operations-helper read only	No	No
Use Import Center	Yes	Yes	No	No	No
Read documents/tasks as assigned staff	Yes	Yes	event-manager/viewer where allowed	No	No
Create audit logs	Only with matching mutation	Only with matching mutation	Only where rules allow matching mutation	Only check-in audit path	No

Firestore Rules Reference

Important helper functions include `isSignedIn`, `isProtectedOwner`, `isApprovedAdmin`, `activeStaffProfile`, `activeStaffAssignment`, `isAssignedScanner`, `canReadTask`, and `canManageTask`.

Rules distinguish `resource.data` from `request.resource.data`, and `create/read/update/delete` paths are intentionally different. Query rules are not filters: if a query can return forbidden documents, Firestore denies the whole query.

Appendix A. Field-Level Firestore Dictionary

This appendix documents fields that can be proven from current source code, validators, Firestore Rules comments, and tests. Where evidence is incomplete, the entry is marked UNKNOWN — REQUIRES FUTURE VERIFICATION.

settings/accessControl

Field	Type	Required	Purpose	Written by	Read by	Permission/security importance	Leq sta
approvedEmails	list	Yes	Secondary organizer allowlist used by authorization checks.	src/services/accessManagementService.js	src/auth/AuthProvider.jsx, src/utils/accessRoles.js, Firestore Rules	High. Organizer authorization depends on normalized membership.	Cui
rolesByEmail	map<string,string>	No	Access-type lookup by organizer email.	src/services/accessManagementService.js	src/utils/accessRoles.js	Medium. UI role labels and capabilities use it.	Cui
approvedOrganizerRecords	map<string,map>	No	Organizer metadata keyed by normalized email.	src/services/accessManagementService.js	src/utils/accessRoles.js, src/pages/SettingsPage.jsx	High. Status, access type, date added, and removal state live here.	Cui
approvedOrganizerRecords.{email}.email	string	Yes when record exists	Canonical organizer email.	src/services/accessManagementService.js	src/pages/SettingsPage.jsx	High. Must match normalized allowlist identity.	Cui
approvedOrganizerRecords.{email}.accessType	string	No	Owner-facing access type label such as admin or organizer.	src/services/accessManagementService.js	src/utils/accessRoles.js, src/pages/SettingsPage.jsx	Medium. Drives capability display.	Cui
approvedOrganizerRecords.{email}.status	string	Yes when record exists	Organizer lifecycle state such as active, disabled, restored, or removed.	src/services/accessManagementService.js	src/utils/accessRoles.js, src/pages/SettingsPage.jsx	High. Only active records should authorize.	Cui
approvedOrganizerRecords.{email}.addedAt	timestamp/string	No	Date the organizer was first added, when available.	src/services/accessManagementService.js	src/pages/SettingsPage.jsx	Low. Audit/display field.	Cui
approvedOrganizerRecords.{email}.updatedAt	timestamp/string	No	Most recent status or metadata change time.	src/services/accessManagementService.js	src/pages/SettingsPage.jsx	Low. Operational history support.	Cui
updatedAt	timestamp	No	Last document-level mutation time.	src/services/accessManagementService.js	Settings diagnostics	Medium. Useful for change review.	Cui

Field	Type	Required	Purpose	Written by	Read by	Permission/security importance	Legacy status
updatedBy	string	No	UID or email of last mutating actor.	src/services/accessManagementService.js	Settings diagnostics	Medium. Audit-adjacent metadata.	Current

settings/integrations

Field	Type	Required	Purpose	Written by	Read by	Permission/security importance	Legacy status
integrations	map<string,map>	Yes	Supported integration states grouped by integration key.	src/services/integrationSettingsService.js	src/pages/SettingsPage.jsx	High. Must remain owner-managed, truthfully displayed, and non-authorizing.	Current
integrations.{key}.status	string	No	Human-readable integration state such as connected, disconnected, disabled, or not configured.	src/services/integrationSettingsService.js	src/pages/SettingsPage.jsx	Medium. Owner-facing operational truth.	Current
integrations.{key}.updatedAt	timestamp	No	Per-integration last update time.	src/services/integrationSettingsService.js	src/pages/SettingsPage.jsx	Low. Review aid.	Current
integrations.{key}.updatedBy	string	No	Actor who changed that integration state.	src/services/integrationSettingsService.js	src/pages/SettingsPage.jsx	Low. Traceability only.	Current
updatedAt	timestamp	No	Last full settings update time.	src/services/integrationSettingsService.js	src/pages/SettingsPage.jsx	Low	Current
updatedBy	string	No	Last mutating actor.	src/services/integrationSettingsService.js	src/pages/SettingsPage.jsx	Low	Current

events/{eventId}

Field	Type	Required	Purpose	Written by	Read by	Permission/security importance	Legacy status
eventId	string	Yes	Stable event document identifier.	src/services/eventService.js	Most event-scoped pages	High. Route and scope anchor.	Current
eventName	string	Yes	Primary event title.	src/services/eventService.js	Shell, Events, Overview, imports, reports	Medium. Working Event visibility depends on it.	Current
eventDate	string/timestamp	Yes	Event date used for sorting and readiness.	src/services/eventService.js	Events, Overview, shell, reports	Medium	Current
status	string	Yes	Event lifecycle state such as draft, registration-open, live, or completed.	src/services/eventService.js	Events, Overview, shell	High. Controls organizer interpretation.	Current
venueName	string	No	Venue label.	src/services/eventService.js	Events, Overview, runbooks	Low	Current

Field	Type	Required	Purpose	Written by	Read by	Permission/security importance	Legacy status
location	string	No	Address or place description.	src/services/eventService.js	Events, Overview	Low	Current
timezone	string	No	Event timezone.	src/services/eventService.js	Event utilities	Medium. Schedule calculations depend on it.	Current
currency	string	No	Default finance currency code.	src/services/eventService.js	Payments, Overview, Operations	Medium	Current
capacity	number	No	Planned guest capacity.	src/services/eventService.js	Overview, review metrics	Low	Current
category	string	No	Event category or type.	src/services/eventService.js	Events, tutorial, capability defaults	Medium	Current
capabilities	map	No	Feature switches/defaults by event.	src/services/eventService.js	Events, route logic, tutorial	High. Rules validate supported shape.	Current
createdAt	timestamp	No	Creation time.	src/services/eventService.js	Events sorting/history	Low	Current
createdBy	string	No	Creator identity.	src/services/eventService.js	Audit context	Low	Current
updatedAt	timestamp	No	Last edit time.	src/services/eventService.js	Events, reports	Low	Current
updatedBy	string	No	Last editor identity.	src/services/eventService.js	Audit context	Low	Current

events/{eventId}/staffAssignments/{uid}

Field	Type	Required	Purpose	Written by	Read by	Permission/security importance	Legacy status
uid	string	Yes	Assigned Firebase UID.	src/services/staffManagementService.js	src/auth/AuthProvider.jsx, Settings	High. Must match document id and auth UID in rules.	Current
eventId	string	Yes	Assigned event scope.	src/services/staffManagementService.js	AuthProvider, Settings	High. Assignment scoping depends on it.	Current
email	string	No	Staff email display/search field.	src/services/staffManagementService.js	Settings	Medium	Current
displayName	string	No	Staff display label.	src/services/staffManagementService.js	Settings	Low	Current
role	string	Yes	Assignment role such as scanner, viewer, event-manager, or operations-helper.	src/services/staffManagementService.js	AuthProvider, route access, Settings	High. Capabilities depend on role.	Current
status	string	Yes	Assignment state, typically active or disabled/removed variant.	src/services/staffManagementService.js	AuthProvider, Settings	High. Only active assignments should authorize.	Current
assignedAt	timestamp	No	When assignment was created.	src/services/staffManagementService.js	Settings history	Low	Current
assignedBy	string	No	Actor creating assignment.	src/services/staffManagementService.js	Settings history	Low	Current

Field	Type	Required	Purpose	Written by	Read by	Permission/security importance	Legacy status
updatedAt	timestamp	No	Last assignment change time.	src/services/staffManagementService.js	Settings	Low	Current
updatedBy	string	No	Last mutating actor.	src/services/staffManagementService.js	Settings	Low	Current

staffProfiles/{uid}

Field	Type	Required	Purpose	Written by	Read by	Permission/security importance	Legacy status
uid	string	Yes	Staff Firebase UID.	src/services/staffManagementService.js	AuthProvider, Settings	High. Must match auth UID in rules.	Current
email	string	Yes	Staff email.	src/services/staffManagementService.js	AuthProvider, Settings	Medium	Current
displayName	string	No	Staff label.	src/services/staffManagementService.js	AuthProvider, Settings	Low	Current
status	string	Yes	Global staff profile state.	src/services/staffManagementService.js	AuthProvider, Settings	High. Inactive profiles should not authorize.	Current
accessType	string	No	Owner-facing classification for staff profile.	src/services/staffManagementService.js	Settings	Low	Current
notes	string	No	Owner-maintained operational note.	src/services/staffManagementService.js	Settings	Low	Current
createdAt	timestamp	No	Creation time.	src/services/staffManagementService.js	Settings	Low	Current
updatedAt	timestamp	No	Last change time.	src/services/staffManagementService.js	Settings	Low	Current

events/{eventId}/tasks/{taskId}

Field	Type	Required	Purpose	Written by	Read by	Permission/security importance	Legacy status
taskId	string	Yes	Stable task identifier.	src/services/taskService.js	Tasks page, dashboard readiness	Medium	Current
eventId	string	Yes	Event scope.	src/services/taskService.js	Tasks, dashboard, audits	High	Current
title	string	Yes	Task name.	src/services/taskService.js	Tasks, dashboard, readiness	Medium	Current
description	string	No	Task detail text.	src/services/taskService.js	Tasks	Low	Current
status	string	Yes	Workflow state.	src/services/taskService.js	Tasks, readiness	High. Rules validate allowed values.	Current
priority	string	No	Priority grouping.	src/services/taskService.js	Tasks, dashboard	Medium	Current
dueDate	string/timestamp	No	Due date.	src/services/taskService.js	Tasks, dashboard	Medium	Current

Field	Type	Required	Purpose	Written by	Read by	Permission/security importance	Legacy status
assigneeName	string	No	Human assignee label.	src/services/taskService.js	Tasks	Low	Current
linkedResourceId	string	No	Resource relationship if created from another module.	Task prefill helpers	Tasks	Medium	Current
linkedRunOfShowId	string	No	Run-of-show relationship if created from another module.	Task prefill helpers	Tasks	Medium	Current
createdAt	timestamp	No	Creation time.	src/services/taskService.js	Tasks	Low	Current
updatedAt	timestamp	No	Last edit time.	src/services/taskService.js	Tasks	Low	Current

registrations/{registrationId}

Field	Type	Required	Purpose	Written by	Read by	Permission/security importance	Legacy status
registrationId	string	Yes	Stable registration identifier.	src/services/registrationService.js, src/services/importService.js	Registrations, payments, tickets, check-in, communications	High	Current
eventId	string	Yes	Event scope.	Registration and import services	Most organizer modules	High. Event scoping and rule queries depend on it.	Current
sourceRowId	string	No	Deterministic import lineage identifier.	src/services/importService.js	Imports, duplicate detection	Medium	Current
fullName	string	Yes	Primary attendee or booking name.	Registration/import services	Registrations, tickets, check-in, communications, audits	Medium	Current
buyerName	string	No	Purchaser label when different from attendee.	Import and registration services	Registrations, payments, communications	Low	Current
attendeeNames	list	No	Group attendee names.	Import helpers	Registrations, communications	Low	Current
email	string	No	Guest contact email.	Registration/import services	Registrations, communications, payments	Medium	Current
phone	string	No	Guest contact phone.	Registration/import services	Registrations, communications	Medium	Current
groupName	string	No	Group booking grouping.	Registration/import services	Registrations, communications	Low	Current
personsAttending	number	Yes	Guest count for the record.	Registration/import services	Overview, registrations, check-in, capacity metrics	High. Summary totals use it.	Current
registrationStatus	string	No	Organizer-facing	Registration service	Registrations, reports	Medium	Current

Field	Type	Required	Purpose	Written by	Read by	Permission/security importance	Legacy status
			registration lifecycle.				
paymentStatus	string	Yes	Finance classification such as paid, pending, complimentary, or door.	Registration/import/finance services	Payments, overview, communications, check-in helper lists	High	Current
amountDue	number	No	Expected amount.	Registration/import services	Payments, reports	Medium	Current
amountPaid	number	No	Recorded amount collected.	Registration/import/finance services	Payments, reports	Medium	Current
paymentMethod	string	No	Payment evidence method.	Registration/import/finance services	Payments, reconciliation	Medium	Current
paymentReference	string	No	Human-entered payment trace id.	Registration/import/finance services	Payments, reconciliation	Medium	Current
ticketStatus	string	No	Ticket lifecycle state.	src/services/ticketService.js	Tickets, check-in, reports	High	Current
ticketCode	string	No	QR-safe ticket code.	src/services/ticketService.js	Tickets, QR, check-in	High. Must remain event-safe and unique.	Current
ticketAssignedAt	timestamp	No	Ticket assignment time.	src/services/ticketService.js	Tickets, audits	Low	Current
ticketAssignedBy	string	No	Ticket assigning actor.	src/services/ticketService.js	Tickets, audits	Low	Current
checkedIn	boolean	No	Attendance state.	src/services/ticketService.js narrow update paths	Check-In, dashboard, reports	High. Scanner path is deliberately narrow.	Current
checkInTime	timestamp	No	Check-in timestamp.	Ticket/check-in service	Check-In, audits	High	Current
checkedInBy	string	No	Actor who completed check-in.	Ticket/check-in service	Check-In, audits	High	Current
historicalAttendanceStatus	string	No	Non-scanner attendance evidence marker.	Historical attendance helpers	Reports/review only	Medium	Legacy-tolerant
notes	string	No	Organizer notes.	Registration/import services	Registrations, reports	Low	Current
createdAt	timestamp	No	Creation time.	Registration/import services	Registrations, reports	Low	Current
updatedAt	timestamp	No	Last update time.	Registration/import/ticket/check-in services	Registrations, reports, audits	Low	Current

auditLogs/{logId}

Field	Type	Required	Purpose	Written by	Read by	Permission/security importance	Legacy status
logId	string	Yes	Stable audit entry id.	Audit-coupled write services	QA, review flows	High. Append-only evidence key.	Current

Field	Type	Required	Purpose	Written by	Read by	Permission/security importance	Legacy status
eventId	string	Yes	Event scope for the change.	All audited write services	QA, reports, troubleshooting	High	Current
targetType	string	Yes	Changed entity type such as registration, operationsLedger, contactLink, runOfShow, or resource.	Audited write services	QA, troubleshooting	High	Current
targetId	string	Yes	Changed entity identifier.	Audited write services	QA, troubleshooting	High	Current
action	string	Yes	Action code such as create, update, delete, assign-ticket, check-in, or undo-check-in.	Audited write services	QA, troubleshooting	High	Current
timestamp	timestamp	Yes	Action time.	Audited write services	QA, troubleshooting	Medium	Current
performedBy	string	Yes	Actor UID/email display token.	Audited write services	QA, troubleshooting	High	Current
details	map	No	Action-specific evidence payload.	Audited write services	QA, troubleshooting	High. Rules validate that details match the mutation.	Current

operationsLedger/{ledgerEntryId}

Field	Type	Required	Purpose	Written by	Read by	Permission/security importance	Legacy status
ledgerEntryId	string	Yes	Stable ledger row id.	src/services/operationsLedgerService.js	Operations, dashboard, reconciliation	High	Current
eventId	string	Yes	Event scope.	Operations service	Operations, dashboard, reconciliation	High	Current
type	string	Yes	Operations classification such as expense, commitment, partner, supplier, or in-kind support.	Operations service	Operations, reports	Medium	Current
status	string	Yes	Row status.	Operations service	Operations, dashboard	Medium	Current
label	string	Yes	Primary row name/description.	Operations service	Operations	Medium	Current
amount	number	No	Monetary amount when applicable.	Operations service	Operations, reconciliation, dashboard	Medium	Current
counterpartyName	string	No	Supplier/partner name.	Operations service	Operations	Low	Current
dueDate	string/timestamp	No	Date-linked operational commitment.	Operations service	Operations	Low	Current
notes	string	No	Owner note field.	Operations service	Operations	Low	Current

Field	Type	Required	Purpose	Written by	Read by	Permission/security importance	Legacy status
createdAt	timestamp	No	Creation time.	Operations service	Operations	Low	Current
updatedAt	timestamp	No	Last change time.	Operations service	Operations, audits	Low	Current

events/{eventId}/documents/{documentId}

Field	Type	Required	Purpose	Written by	Read by	Permission/security importance	Legacy status
documentId	string	Yes	Stable document reference id.	src/services/documentService.js	Documents page, linked workflows	Medium	Current
eventId	string	Yes	Event scope.	Document service	Documents	High	Current
title	string	Yes	Reference title.	Document service	Documents	Medium	Current
category	string	No	Document category.	Document service	Documents filters	Medium	Current
status	string	No	Document status.	Document service	Documents	Medium	Current
url	string	Yes	External or internal reference URL.	Document service	Documents	High. URL validation protects integrity.	Current
expiresOn	string/timestamp	No	Expiry/review date.	Document service	Documents	Low	Current
linkedTaskId	string	No	Task relationship.	Document service/task prefill	Documents, tasks	Medium	Current
notes	string	No	Operational note.	Document service	Documents	Low	Current
createdAt	timestamp	No	Creation time.	Document service	Documents	Low	Current
updatedAt	timestamp	No	Last update time.	Document service	Documents	Low	Current

events/{eventId}/runOfShow/{itemId}

Field	Type	Required	Purpose	Written by	Read by	Permission/security importance	Legacy status
itemId	string	Yes	Stable run item id.	src/services/runOfShowService.js	Run of Show, dashboard	Medium	Current
eventId	string	Yes	Event scope.	Run of Show service	Run of Show, dashboard	High	Current
title	string	Yes	Sequence item label.	Run of Show service	Run of Show, dashboard	Medium	Current
startTime	string	Yes	Planned start time.	Run of Show service	Run of Show	High. Rules require valid timing shape.	Current
endTime	string	No	Planned end time.	Run of Show service	Run of Show	Medium	Current
status	string	Yes	Operational state.	Run of Show service	Run of Show, dashboard	Medium	Current

Field	Type	Required	Purpose	Written by	Read by	Permission/security importance	Legacy status
location	string	No	On-day location.	Run of Show service	Run of Show	Low	Current
ownerName	string	No	Human owner/operator.	Run of Show service	Run of Show	Low	Current
dependencyIds	list	No	Sequence dependencies.	Run of Show service	Run of Show	Medium	Current
linkedResourceIds	list	No	Resource relationships.	Run of Show service	Run of Show, resources	Medium	Current
notes	string	No	Event-day note.	Run of Show service	Run of Show	Low	Current
createdAt	timestamp	No	Creation time.	Run of Show service	Run of Show	Low	Current
updatedAt	timestamp	No	Last update time.	Run of Show service	Run of Show	Low	Current

events/{eventId}/resources/{resourceId}

Field	Type	Required	Purpose	Written by	Read by	Permission/security importance	Legacy status
resourceId	string	Yes	Stable resource record id.	src/services/eventResourceService.js	Resources, dashboard, run-of-show links	Medium	Current
eventId	string	Yes	Event scope.	Resource service	Resources, dashboard	High	Current
name	string	Yes	Resource label.	Resource service	Resources	Medium	Current
category	string	No	Resource type or grouping.	Resource service	Resources filters	Medium	Current
status	string	Yes	Packing/return/availability state.	Resource service	Resources, dashboard	Medium	Current
quantityRequired	number	No	Planned quantity.	Resource service	Resources	Low	Current
quantityConfirmed	number	No	Confirmed quantity.	Resource service	Resources	Low	Current
supplierName	string	No	Vendor/source label.	Resource service	Resources	Low	Current
pickupDate	string/timestamp	No	Pickup timing.	Resource service	Resources	Low	Current
returnDate	string/timestamp	No	Return timing.	Resource service	Resources	Low	Current
linkedTaskId	string	No	Task relationship.	Resource service/task prefill	Resources, tasks	Medium	Current
notes	string	No	Resource note.	Resource service	Resources	Low	Current
createdAt	timestamp	No	Creation time.	Resource service	Resources	Low	Current
updatedAt	timestamp	No	Last update time.	Resource service	Resources	Low	Current

contacts/{contactId},
organizations/{organizationId},
events/{eventId}/contactLinks/{linkId}

Field	Type	Required	Purpose	Written by	Read by	Permission/security importance	Legacy status
contactId / organizationId /	string	Yes	Stable id for the record	src/services/contactService.js	Contacts page	Medium	Current

Field	Type	Required	Purpose	Written by	Read by	Permission/security importance	Legacy status
linkId			type.				
fullName / organizationName	string	Yes	Main display label.	Contact service	Contacts, links, tasks prefills	Medium	Current
email	string	No	Contact email.	Contact service	Contacts, relationships	Medium	Current
phone	string	No	Contact phone.	Contact service	Contacts	Medium	Current
category	string	No	Contact or organization classification.	Contact service	Contacts filters	Low	Current
notes	string	No	Owner note field.	Contact service	Contacts	Low	Current
eventId	Yes on link	Event-scoped relationship target.	Contact service	Contacts page	High. Link scope must match event.	Current	
contactId / organizationId on link	string	One of them required	Target record relationship.	Contact service	Contacts page	High. Relationship integrity depends on it.	Current
roleLabel	string	No	Event-specific role such as sponsor, vendor, speaker, or venue contact.	Contact service	Contacts page	Medium	Current
status	string	No	Relationship activity state.	Contact service	Contacts page	Medium	Current
createdAt	timestamp	No	Creation time.	Contact service	Contacts page	Low	Current
updatedAt	timestamp	No	Last update time.	Contact service	Contacts page	Low	Current

accessRequests/{requestId}

Field	Type	Required	Purpose	Written by	Read by	Permission/security importance	Legacy status
requestId	string	Yes	Access request id.	src/services/accessRequestContract.js workflow	Access tooling	Medium	Current
email	string	Yes	Requesting email.	Access request flow	Future access-review flow	High. Identity evidence.	Current
displayName	string	No	Human requester label.	Access request flow	Future review flow	Low	Current
reason	string	No	Why access is requested.	Access request flow	Future review flow	Low	Current
status	string	Yes	Request lifecycle state.	Access request flow or admin tooling	Future review flow	Medium	Current
createdAt	timestamp	No	Request time.	Access request flow	Future review flow	Low	Current

History collections

`settings/accessControl/history/{historyId}`,
`settings/integrations/history/{historyId}`,
`staffHistory/{historyId}`, and
`events/{eventId}/staffAssignmentHistory/{historyId}`
are append-only evidence logs. The
exact shape varies by change type, but
each should at minimum carry actor
identity, changed entity, changed-at
time, previous/next status or access
value, and a reason or note when
provided.

Appendix B.

Command Safety Reference

Use this appendix before running any script that touches Firebase, emulator data, or owner-managed records.

Command	Purpose	Safety class	Data target	Prerequisites	Expected result	Common failure
<code>npm run docs:generate</code>	Render HTML and PDF manual outputs.	LOCAL FILE WRITE	docs/generated, owner-copy PDF folder	Node, Python PDF helper path, repo source docs	New HTML/PDF generated	Markdown/diagram/PDF helper failure
<code>npm run docs:validate</code>	Validate required docs source and generated artifacts.	READ-ONLY	Repository files	Generated docs exist	Pass message with file count	Missing appendix/runbook/manual file
<code>npm run docs:pdf-check</code>	Inspect PDF structure and render validation images.	LOCAL FILE WRITE	output/pdf-validation/phase-3b	Generated PDF, pypdf, pypdfium2	Bookmark/text/render checks pass and page PNGs exist	Missing helper dependency, raw syntax still visible
<code>npm run lint</code>	Static quality checks.	READ-ONLY	Repository source	Dependencies installed	Exit code 0	ESLint violation
<code>npm test</code>	Unit/source-contract regression checks.	READ-ONLY	Local code only	Dependencies installed	All expected tests pass/skip	Source regression, date-sensitive failure
<code>npm run build</code>	Production bundle build.	LOCAL BUILD OUTPUT	dist/	Dependencies installed	Bundle generated	Vite import/config failure
<code>npm run product:routes</code>	Route inventory contract check.	READ-ONLY	Local code only	Node deps installed	Route count and labels pass	Route mismatch
<code>npm run product:docs</code>	Product documentation guardrails.	READ-ONLY	Local docs only	Docs current	Docs check passes	Stale language or missing docs
<code>npm run product:qa</code>	Full local QA wrapper including emulators and smoke.	EMULATOR WRITE	Local auth/firestore emulators only	Java runtime, free ports 8080/9099, installed browsers	All QA phases pass	Port collision, emulator startup failure, smoke E2E failure
<code>npm run doctor:changed</code>	Changed-scope React Doctor review.	READ-ONLY	Local code only	Clean Git base and installed deps	No issues found	Changed-scope React issue
<code>npm run e2e:smoke</code>	Chromium smoke test under emulators.	EMULATOR WRITE	Local emulators only	Free ports, Playwright browsers	Smoke route pass	Emulator startup or navigation failure
<code>npm run e2e:full</code>	Full Chromium E2E suite under emulators.	EMULATOR WRITE	Local emulators only	Same as smoke plus stable local runner	All E2E tests pass	Runner EPIPE, lingering ports, feature regression
<code>npm run admin:verify-production-fixtures</code>	Read-only confirmation that expected production fixtures exist.	PRODUCTION READ	Production Firestore	Safe credentials/session and explicit reason	Fixture report only	ADC or permission issue
<code>npm run admin:verify-</code>	Read-only production	PRODUCTION READ	Production Firestore	Same as above	Count report only	Wrong project, auth failure

Command	Purpose	Safety class	Data target	Prerequisites	Expected result	Common failure
production-counts	record counting.					
npm run admin:ensure-access	Administrative access-control maintenance helper.	ADMINISTRATIVE - VERIFY BEFORE RUNNING	Production settings/accessControl unless redirected	Explicit approval and backup evidence	Access-control write completes	Wrong target project, protected owner risk
npm run admin:replace-codex-test-with-demo	Historical admin maintenance helper for demo-event migration.	ADMINISTRATIVE - VERIFY BEFORE RUNNING	Firestore records	Exact dry-run understanding	Intended migration only	Wrong event scope
npm run firebase:deploy-hosting	Deploy Hosting only.	PRODUCTION DEPLOYMENT	Production Hosting	Built app, approved release plan	Hosting release completes	Wrong project, stale assets, broken build
npm run firebase:deploy-rules	Deploy Firestore Rules and indexes.	PRODUCTION DEPLOYMENT	Production Firestore Rules/indexes	Emulator rule tests, approval, exact diff review	Rules/index deployment completes	Authorization regression or wrong project
npm run firebase:deploy-all	Deploy Hosting plus Rules/indexes.	PRODUCTION DEPLOYMENT	Multiple production Firebase targets	Strong approval and validated all targets	All targets deploy cleanly	Mixed-target regression

Quick Safety Rules

- **READ-ONLY:** safe to run against the repository or production only when the command itself does not mutate state.
- **LOCAL FILE WRITE:** changes files in the working tree or generated artifact folders only.
- **LOCAL BUILD OUTPUT:** writes generated app assets only.
- **EMULATOR WRITE:** can seed, mutate, or clear local emulator data; never production.
- **PRODUCTION READ:** reads live Firebase state without mutating it.
- **ADMINISTRATIVE - VERIFY BEFORE RUNNING:** potentially mutates owner-managed settings or maintenance records and requires narrow intent.
- **PRODUCTION DEPLOYMENT:** deploys live infrastructure. Never run casually.

Appendix C.

Route and Feature Indexes

Route Index

Route	Main purpose	Primary page module	Main supporting services/utilities	Related tests
/dashboard	Event overview, readiness, and next actions	src/pages/DashboardPage.jsx	eventReadiness, registrationMetrics, operationsLedgerService, runOfShowService, eventResourceService	tests/page-organization-pass2-compactness.test.js, dashboard tests
/events	Event planning and Working Event control	src/pages/EventsPage.jsx	eventService, eventPlanning	tests/event-utils.test.js, category/capability tests
/tasks	Event-scoped task management	src/pages/TasksPage.jsx	taskService, taskWorkflow	tests/task-workflow-rules.test.js
/registrations	Guest records and attendance context	src/pages/RegistrationsPage.jsx	registrationService, validators, registrationMetrics	registration tests
/payments	Registration-linked payments	src/pages/PaymentsPage.jsx	financeUtils, paymentStatus	payments boundary tests
/payments/reconciliation	Workbook comparison and review-only reconciliation	src/pages/PaymentReconciliationPage.jsx	reconciliationReadService, paymentReconciliation, reconciliationWorkbook	reconciliation tests
/imports	Import Center workflow	src/pages/ImportsPage.jsx	importService, importUtils, xlsxImport	import-center tests
/tickets	Ticket assignment and QR preparation	src/pages/TicketsPage.jsx	ticketService, ticketUtils, qrTicketUtils	ticketing and QR tests
/check-in	Assisted event-day attendance operations	src/pages/CheckInPage.jsx	ticketService, checkInUtils, eventDayHelpers	check-in rules/tests
/scanner	Assigned-event scanner route	src/pages/ScannerPage.jsx	QrScannerPanel, qrTicketUtils	camera/check-in tests
/operations	Ledger, commitments, partners, and in-kind support	src/pages/OperationsPage.jsx	operationsLedgerService	operations tests
/run-of-show	Event-day sequence planning	src/pages/RunOfShowPage.jsx	runOfShowService, runOfShow	run-of-show tests
/resources	Equipment and supplies tracking	src/pages/ResourcesPage.jsx	eventResourceService, eventResources	resource tests
/documents	Event document references	src/pages/DocumentsPage.jsx	documentService, documentRegistry	document tests

Route	Main purpose	Primary page module	Main supporting services/utilities	Related tests
/contacts	Contact, organization, and relationship directory	src/pages/ContactsPage.jsx	contactService, contactDirectory	contact tests
/event-review	Reports and read-only review surfaces	src/pages/EventReviewPage.jsx	review helpers, finance/report utils	event review tests
/communications	Copy-only message preparation	src/pages/CommunicationsPage.jsx	communicationsUtils	communications tests
/settings	Owner/admin settings surfaces	src/pages/SettingsPage.jsx	access/staff/integration services	settings tests
/qa	System QA diagnostics	src/pages/QaPage.jsx	runtimeHealth, qaHelper	production QA tests

Feature-to-File Index

Feature	Main UI files	Main service files	Main utilities	Rule/test anchors
Organizer access	src/pages/SettingsPage.jsx	src/services/accessManagementService.js, src/services/staffManagementService.js	src/utils/accessRoles.js	firestore.rules access-control/staff blocks; settings tests
Registration workflow	src/pages/RegistrationsPage.jsx	src/services/registrationService.js	src/utils/validators.js, src/utils/registrationMetrics.js	registrations rules and metrics tests
Payments	src/pages/PaymentsPage.jsx	registration service bulk finance paths	src/utils/financeUtils.js, src/utils/paymentStatus.js	payments boundary tests
Ticketing and QR	src/pages/TicketsPage.jsx, src/components/tickets/TicketQrCode.jsx	src/services/ticketService.js	src/utils/ticketUtils.js, src/utils/qrTicketUtils.js	ticket/QR tests
Check-In and scanner	src/pages/CheckInPage.jsx, src/pages/ScannerPage.jsx	src/services/ticketService.js	src/utils/checkInUtils.js	camera/check-in/rules tests
Import Center	src/pages/ImportsPage.jsx	src/services/importService.js	src/utils/importUtils.js, src/utils/xlsxImport.js	import-center tests
Operations	src/pages/OperationsPage.jsx	src/services/operationsLedgerService.js	operations workspace helpers	operations tests
Run of Show	src/pages/RunOfShowPage.jsx	src/services/runOfShowService.js	src/utils/runOfShow.js	run-of-show tests
Resources	src/pages/ResourcesPage.jsx	src/services/eventResourceService.js	src/utils/eventResources.js	resource tests
Documents	src/pages/DocumentsPage.jsx	src/services/documentService.js	src/utils/documentRegistry.js	document tests
Contacts	src/pages/ContactsPage.jsx	src/services/contactService.js	src/utils/contactDirectory.js	contact tests
Tutorial/onboarding	src/tutorial/*	onboarding preference reads/writes	tutorial registries/helpers	tutorial tests

Appendix D. Error Index and Search Keywords

Error Index

Message or symptom	Most likely area	First runbook
Failed to fetch dynamically imported module	Hosting deploy drift or browser cache	Application and Session Failures
Missing or insufficient permissions	Firestore Rules or organizer/staff access mismatch	Permission and Access Failures
Redirect loop to /login	Authentication state or access resolver	Application and Session Failures
Scanner opens but cannot find assigned event	Staff assignment or Working Event mismatch	Scanner, Ticket, and Check-In Failures
Ticket code collision	Ticket generation/import data	Import and Data Repair
Firestore rules are not filters	Query shape exceeds rule visibility	Permission and Access Failures
Emulator ports already in use	Local QA infrastructure	Development, Build, and Test Failures
NO_ADC_FOUND or Firebase admin credential issue	Production-read admin tooling	Deployment, Firebase, and Project Targeting
Playwright smoke timeout	Local app boot, route error, or emulator startup	Development, Build, and Test Failures
React Doctor warning	Changed code health issue	Development, Build, and Test Failures

Search Keywords

- dynamic import stale chunk deployment cache-bust
- permission denied approved organizer protected owner staff assignment
- scanner camera QR html5-qr code GSV:TICKET
- ticket code collision duplicate registration

import

- Firestore rules are not
filters query denied
- firebase project
selected wrong hosting
rules deploy
- emulator port 8080 9099
collision
- React Doctor changed
scope warning
- worktree detached branch
local changes lost
- documentation generation
pdf bookmark toc

Appendix E.

Documentation

Maintenance

Procedure

When this manual must be updated

- Any meaningful runtime change that affects routes, workflows, permissions, settings, schemas, statuses, calculations, imports, reports, or repair procedures.
- Any change to Firestore Rules, indexes, or admin service write shapes.
- Any UI reorganization that changes where operators find key controls.
- Any change that invalidates a screenshot, diagram, command reference, or runbook.

Required maintenance actions

1. Update the affected source Markdown files first.
2. Regenerate diagrams if any flow, route, or permission relationship changed.
3. Replace screenshots when the visible interface, labels, or workflow steps changed.
4. Regenerate HTML and PDF:
`npm run docs:generate`
5. Validate sources and final PDF: `npm run docs:validate` `npm run docs:pdf-check`
6. Run repository safety checks appropriate to the change:
`npm run lint` `npm test` `npm run build` `npm run product:routes` `npm run product:docs`
7. When docs tooling changes, also run: `npm run product:qa` `npm run doctor:changed`
8. Replace the owner copy only after the final PDF passes

visual and structural validation.

Screenshot replacement rule

- Replace screenshots when labels, layout, tabs, buttons, or critical data panels changed.
- Prefer CODEX_DEMO, local emulator data, or other non-sensitive synthetic records.
- Do not include passwords, tokens, private guest data, or `.env.local` values.
- Record filename, route, date captured, and redaction status in the screenshot catalogue.

Diagram regeneration rule

- Regenerate diagrams whenever routes, auth flow, permissions, deployment workflow, import flow, QR flow, or troubleshooting flow changes.
- Keep editable Mermaid source under `docs/diagrams/`.
- Confirm the rendered SVG/PNG is readable at normal PDF zoom.

Bookmark and link validation rule

- Every regenerated manual must have a non-zero bookmark count.
- TOC entries and major internal links must be checked during `docs:pdf-check`.
- If bookmarks disappear, treat the PDF as not releasable.

Changelog rule

- Add a documentation changelog entry whenever the manual structure, rendering pipeline, appendices, screenshots, or runbooks change materially.
- State whether runtime app behavior changed. For

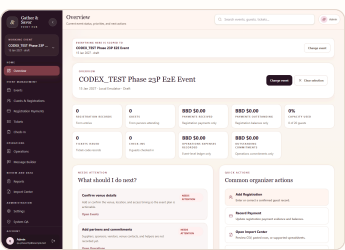
documentation-only work,
state that it did not.

Screenshot Catalog

This catalogue tracks screenshots intentionally included in the technical manual. Only synthetic, redacted, or otherwise safe captures belong here.

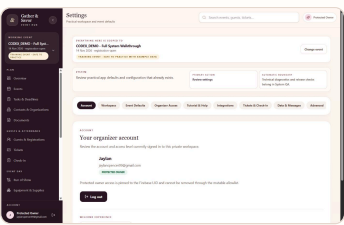
Filename	Date captured	Route/interface	Data source	Redaction status	Manual section	Replacement policy
overview-emulator.png	2026-08-21 or earlier archived safe capture	/dashboard	Local emulator synthetic event	Synthetic only	Application Overview	Replace when layout, labels, or workflow materially change
settings-organizer-access.png	2026-08-21 or earlier archived safe capture	/settings	CODEX_DEMO production-safe synthetic event	Protected Owner visible; no secrets	Settings and staff access	Replace when layout, labels, or workflow materially change
import-center-codex-demo.png	2026-08-21 or earlier archived safe capture	/imports	CODEX_DEMO production-safe synthetic event	Synthetic only	Imports, Exports, and QR Systems	Replace when layout, labels, or workflow materially change

Overview workspace



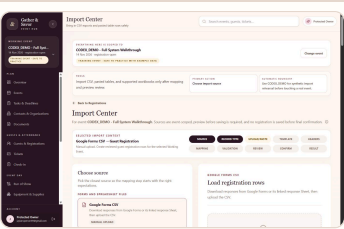
Caption: Overview workspace. Route: /dashboard. Data source: Local emulator synthetic event.

Settings workspace and organizer account view



Caption: Settings workspace and organizer account view. Route: /settings. Data source: CODEX_DEMO production-safe synthetic event.

Import Center staged workflow



Caption: Import Center staged workflow. Route: /imports. Data source: CODEX_DEMO production-safe synthetic event.

Problem and Repair Register

Problem ID	Symptom	Root cause	Date first seen	Date last seen	Occurrences	Affected feature	Affected files	Repair	Regression test	Runbook
GSV-PROB-001	Document helper test expected one expired document but current date made two expired.	Date-sensitive fixture used 2026-08-20 and summary helper used current clock.	2026-08-21	2026-08-21	1 known	Documents/tests	tests/document-contact-foundation.test.js	Changed fixture to 2026-08-30 to preserve expiring-soon intent.	npm test	Build/deployment recovery; test manual
GSV-PROB-002	product:qa emulator startup failed because ports were taken.	Concurrent emulator-backed smoke run used auth/firestore ports.	2026-08-21	2026-08-21	1 known	QA runner	package.json; scripts/product/runProductCommand.mjs; playwright.config.js	Reran product:qa serially after smoke completed.	npm run product:qa	Build/deployment recovery
GSV-PROB-003	Generated dependency cleanup inside archive could not run.	Runtime command policy blocked recursive delete before execution.	2026-08-21	2026-08-21	1 known	Development environment cleanup	Archived migration evidence	Preserved material and documented reclaimable follow-up.	N/A	Legacy system
GSV-PROB-004	Permission denied while UI appears available.	Potential drift between frontend access roles and Firestore Rules.	Unknown	Unknown	Unknown	Permissions	src/utils/accessRoles.js; firestore.rules	Compare route gate, user access source, target document fields, and rule tests.	npm run product:qa	Permission Failure Runbooks

Documentation

Changelog

2026-08-21

- Created Phase 3 technical manual source structure.
- Added Firestore data dictionary, permission matrix, runbooks, ADRs, problem register, templates, diagrams, and PDF generation.
- Added permanent documentation update policy to AI agent governance.

2026-08-21 Phase 3B

- Repaired the documentation rendering architecture so Markdown, tables, and Mermaid are rendered instead of printed as raw source.
- Expanded the manual with grouped repair runbooks, command-safety appendix, route and feature indexes, and a field-level Firestore dictionary appendix.
- Added PDF structural validation, bookmark post-processing, page rendering, and contact-sheet generation for visual QA.
- Replaced the owner-current manual copy and preserved the older 24-page Phase 3 PDF as superseded history.
- No runtime app behavior changed. No Firebase targets were deployed.